Chapter 5

Constrained Optimization

5.1 Lagrange Multipliers

One of the most common problems in calculus is that of optimizing a function, but it is often difficult to find a closed form for the function that must be optimized. Such difficulties frequently arise when one wishes to minimize or maximize a function subject to certain outside conditions or constraints. The method of Lagrange multipliers is a powerful tool for solving this class of problems without the need to explicitly solve the conditions and use them to eliminate extra variables.

Let $f, h_i \colon \mathbb{R}^n \to \mathbb{R}$ for $1 \le i \le m$ with $m \le n$. In this section we consider a problem of the form

$$\text{Minimize } f(x) \text{ subjected to } h_i(x) = 0 \text{ for } 1 \le i \le m, \,\&\, m \le n. \tag{5.1}$$

Note that the problem (5.1) can be reformulated as

$$\text{Minimize } f(x) \text{ subjected to } h(x) = 0 \tag{5.2}$$

where $f \colon \mathbb{R}^n \to \mathbb{R}$ and $h \colon \mathbb{R}^n \to \mathbb{R}^m$ with $h(x) = (h_1(x), \dots, h_m(x))$ and $m \le n$.

Here f is called the objective function and h is called the constraint.

Definition 5.1.1. Let $h \colon \mathbb{R}^n \to \mathbb{R}^m$. The set

$$S := \{x \in \mathbb{R}^n : h(x) = c\}$$

is called the level set.

Example 5.1.1. (i) For $n = 2$ and $m = 1$, let us consider $h(x) = x_1^2 + x_2^2$. Then the level set $S := \{x \in \mathbb{R}^2 : h(x) = 1\}$ is the circle of radius 1 centered at the origin.
(ii) For $n = 2$ and $m = 2$, let us consider $h_1(x) = x_1^2 + x_2^2 - 1$ and $h_2(x) = x + y - 1$. Then the level set $S := \{x \in \mathbb{R}^2 : h(x) = 0\}$ is set of two points $\{(1, 0), (0, 1)\}$.
(iii) For $n = 3$ and $m = 2$, let us consider $h_1(x) = x_1^2 + x_2^2 + x_3^2 - 1$ and $h_2(x) = x_3$. Then the level set $S := \{x \in \mathbb{R}^3 : h(x) = 0\}$ is the circle $x_1^2 + x_2^2 = 1$ in the xy-plane in $\mathbb{R}^3$.

$\Diamond$

Exercise 5.1.2. Find the level sets of the following functions:
(i) $h = (h_1, h_2) = 0$ where $h_1(x_1, x_2, x_3) = 3x_1 + 2x_2 - x_3 - 5$, $h_2(x_1, x_2, x_3) = x_1 - 2x_2 + 3x_3 - 1$
(ii) $h = (h_1, h_2) = 0$ where $h_1(x_1, x_2, x_3) = x_1^2 + x_2^2 + x_3^2 - 1$, $h_2(x_1, x_2, x_3) = x_1 + x_2 + x_3$

‡

Definition 5.1.2. Any point satisfying the constraint of the problem (5.2) is called a feasible point. That is, the set of feasible points are $S := \{x \in \mathbb{R}^n : h(x) = 0\}$

Definition 5.1.3. A point $x^* \in \mathbb{R}^n$ such that $h(x^*) = 0$ is said to a regular point of h if $\nabla h_1(x^*)$, $\nabla h_2(x^*)$, ..., $\nabla h_m(x^*)$ are linearly independent. This is same as saying x^* is regular point of h if

$$Dh(x^*) = \begin{pmatrix} Dh_1(x^*) \\ \vdots \\ Dh_m(x^*) \end{pmatrix} = \begin{pmatrix} \nabla h_1(x^*)^T \\ \vdots \\ \nabla h_m(x^*)^T \end{pmatrix}$$

has full rank m,

If all points in the level set S are regular points of h then S is called a regular surface and its dimension is $n - m$.

Example 5.1.3. (i) For $n = 2$ and $m = 1$, let us consider $h(x) = x_1^2 + x_2^2 - 4$. Then the level set $S := \{x \in \mathbb{R}^2 : h(x) = 0\}$ is the circle of radius 2 centered at the origin. $\nabla h(x) = (2x, 2y) \neq 0$ on S. Hence S is a regular surface of dimension $2 - 1 = 1$.
(ii) For $n = 2$ and $m = 2$, let us consider $h_1(x) = x_1^2 + x_2^2 - 1$ and $h_2(x) = x_1 + x_2 - 1$. Then the level set $S := \{x \in \mathbb{R}^2 : h(x) = 0\}$ is set of two points $\{(1, 0), (0, 1)\}$. Note that $Dh(x_1, x_2) = \begin{pmatrix} 2x_1 & 2x_2 \\ 1 & 1 \end{pmatrix}$. Let $x^* = (1, 0) \in S$, Then $Dh(x^*) = \begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix}$ has rank 2. Hence $(1, 0)$ is a regular point. Similarly $(0, 1)$ is also a regular point.
(iii) For $n = 3$ and $m = 2$, let us consider $h_1(x) = x_1^2 + x_2^2 + x_3^2 - 1$ and $h_2(x) = x_3$. Then the level set $S := \{x \in \mathbb{R}^3 : h(x) = 0\}$ is the circle $x_1^2 + x_2^2 = 1$ in the xy-plane in $\mathbb{R}^3$. For any point $x^* = (x_1, x_2, 0) \in S$, one of x_1, x_2 is non zero. Hence $Dh(x^*) = \begin{pmatrix} 2x_1 & 2x_2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ has rank 2. Thus S is a regular surface of dim $3 - 2 = 1$.

$\diamondsuit$

Definition 5.1.4. Let $S = \{x \in \mathbb{R}^n : h(x) = 0\}$ be a surface. The tangent space at a point p on the surface S is the set

$$T_pS := \{y \in \mathbb{R}^n : Dh(p)y = 0\}. \tag{5.3}$$

Note that T_pS is a subspace of $\mathbb{R}^n$ and it passes through the origin. If p is a regular point on S then the dimesion of T_pS is $n - m$.

It is often convinient to visualize the tangent space as the plane passing through the point p which we can write as

$$\boxed{TP_pS = p + T_pS = \{p + x : x \in T_pS\}.} \tag{5.4}$$

We call TP_pS as affine tangent space.

Example 5.1.4. Let $h_1(x) = x_1 + 2x_2 - 2$ and $h_2(x) = x_1 - 2x_3$. Then

$$S := \{x \in \mathbb{R}^3 : h_1(x) = 0, h_2(x) = 0\} = \{x \in \mathbb{R}^3 : x_2 + x_3 = 1 \;\&\; x_1 = 2x_3\}$$

which is a line.

Let $p = (0, 1, 0) \in S$. $Dh(p) = \begin{pmatrix} 1 & 2 & 0 \\ 1 & 0 & -2 \end{pmatrix}$. Clearly rank of $Dh(p)$ is 2 and hence p is a regular point. In fact S is regular.

$$\begin{aligned} T_pS &= \left\{ (y_1, y_2, y_3) \in \mathbb{R}^3 : \begin{pmatrix} 1 & 2 & 0 \\ 1 & 0 & -2 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = 0 \right\} \\ &= \{(y_1, y_2, y_3) \in \mathbb{R}^3 : y_1 + 2y_2 = 0, y_1 - 2y_3 = 0\} \\ &= \{(y_1, y_2, y_3) \in \mathbb{R}^3 : y_1 = 2y_3, y_2 = -y_3\} \\ &= \{(2\alpha, -\alpha, \alpha) : \alpha \in \mathbb{R}\} \end{aligned}$$

$\diamondsuit$

Example 5.1.5. Consider $\mathbb{R}^4$ with coordinates (x, y, z, w). Define two scalar functions

$$F_1(x, y, z, w) = x^2 + y^2 + z^2 + w^2 - 1, \qquad F_2(x, y, z, w) = x + y + z + w.$$

The set

$$S = \{(x, y, z, w) \in \mathbb{R}^4 : \ F_1(x, y, z, w) = 0, \ F_2(x, y, z, w) = 0\}$$

is the intersection of the unit 3-sphere S^3 and the hyperplane $x + y + z + w = 0$. Under the usual regularity condition (the gradients $\nabla F_1, \nabla F_2$ are linearly independent at a point), S is a smooth 2-dimensional submanifold of $\mathbb{R}^4$.

Look for a point of the form $p = (a, a, a, -3a)$. The hyperplane condition $x + y + z + w = 0$ is satisfied, and the unit-norm condition $x^2 + y^2 + z^2 + w^2 = 1$ gives

$$3a^2 + 9a^2 = 12a^2 = 1 \quad \Rightarrow \quad a = \frac{1}{\sqrt{12}} = \frac{1}{2\sqrt{3}}.$$

Thus a convenient point on S is

$$p = \left(\tfrac{1}{2\sqrt{3}}, \tfrac{1}{2\sqrt{3}}, \tfrac{1}{2\sqrt{3}}, -\tfrac{3}{2\sqrt{3}}\right).$$

For an implicitly defined manifold $F_1 = 0, F_2 = 0$, the tangent space T_pS is the set of vectors $v = (v_1, v_2, v_3, v_4) \in \mathbb{R}^4$ satisfying the linear conditions

$$\nabla F_1(p) \cdot v = 0, \qquad \nabla F_2(p) \cdot v = 0.$$

Compute the gradients:

$$\nabla F_1(x, y, z, w) = (2x, 2y, 2z, 2w), \qquad \nabla F_2 = (1, 1, 1, 1).$$

At p we therefore require

$$(2p) \cdot v = 0 \quad \Longleftrightarrow \quad p \cdot v = 0, \qquad (1, 1, 1, 1) \cdot v = 0.$$

Writing these out explicitly (and using that the common scalar $a \neq 0$ cancels) gives the equivalent linear system

$$v_1 + v_2 + v_3 - 3v_4 = 0, \qquad v_1 + v_2 + v_3 + v_4 = 0.$$

Subtract the two equations to eliminate the sum $v_1 + v_2 + v_3$:

$$\big(v_1 + v_2 + v_3 + v_4\big) - \big(v_1 + v_2 + v_3 - 3v_4\big) = 4v_4 = 0 \quad \Rightarrow \quad v_4 = 0.$$

Substituting $v_4 = 0$ into either equation yields $v_1 + v_2 + v_3 = 0$. Hence

$$T_pS = \{v \in \mathbb{R}^4 : \ v_4 = 0, \ v_1 + v_2 + v_3 = 0\}.$$

This is a 2-dimensional linear subspace. A convenient basis is

$$u^{(1)} = (1, -1, 0, 0), \qquad u^{(2)} = (1, 0, -1, 0),$$

since each basis vector satisfies $v_4 = 0$ and $v_1 + v_2 + v_3 = 0$.

The tangent plane (affine 2-plane) at the point p is

$$p + T_pS = \left\{ p + \alpha\, u^{(1)} + \beta\, u^{(2)} : \alpha, \beta \in \mathbb{R} \right\}.$$

Equivalently, the tangent plane can be written as the intersection of the two linear equations

$$\nabla F_1(p) \cdot (x - p) = 0, \qquad \nabla F_2(p) \cdot (x - p) = 0,$$

or explicitly

$$p \cdot (x - p) = 0, \qquad (1, 1, 1, 1) \cdot (x - p) = 0.$$

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Example 5.1.6 (Tangent Plane to a graph surface). Let $z = f(x, y)$ be a surface S. Then S can be described as

$$S := \{(x, y, z) \in \mathbb{R}^3 : h(x, y, z) = z - f(x, y) = 0\}.$$

Let $(x_0, y_0, z_0 = f(x_0, y_0))$ be a point on the surface. Then

$$Dh(x_0, y_0, z_0) = \left[-\frac{\partial f}{\partial x} \; -\frac{\partial f}{\partial y}, 1\right].$$

$$T_pS = \left\{(x, y, z) \in \mathbb{R}^3 : z - x\frac{\partial f}{\partial x} - y\frac{\partial f}{\partial y} = 0\right\}$$

One can also show that

$$TP_pS = \left\{(x, y, z) \in \mathbb{R}^3 : (z - z_0) - (x - x_0)\frac{\partial f}{\partial x} - (y - y_0)\frac{\partial f}{\partial y} = 0\right\}$$

$\Diamond$

Definition 5.1.5. A curve γ on a surface S is a function $\gamma\colon (a, b) \to S$.

Theorem 5.1.1. *Let p be a regular point of a surface S and T_pS is the tangent space at p to S. Then $v \in T_pS$ if and only if there exists a curve $\gamma : (-\varepsilon, \varepsilon) \to S$ with $\gamma(0) = p$ and $\gamma'(0) = v$.*

In view of the above theorem, one can define T_pS as set of all $\gamma'(0)$ where $\gamma : (-\varepsilon, \varepsilon) \to S$ is any curve on S with $\gamma(0) = p$.

Proof. Let $\gamma(t)$ be a curve in S such that $\gamma(t_0) = p$ for some t_0 in the domain (a, b) of γ and that $\gamma'(t_0) = v$. Then

$$h(\gamma(t)) = 0 \text{ for all } t \in (a, b).$$

Differentiating both sides, we get

$$\frac{d}{dt}[h(\gamma(t))] = Dh(\gamma(t))\gamma'(t) = 0.$$

Thus at t_0, we have

$$Dh(\gamma(t_0))\gamma'(t_0) = Dh(p)v = 0$$

Hence $v \in T_pS$.

The converse follows from the Implicit Function Theorem. $\square$

5.2 Lagrange Conditions

First of all let us consider a minimization problem of a function of two variables,

$$\text{minimize} f(x, y) \text{ subjected to } g(x, y) = 0. \tag{5.5}$$

Let $p = (a, b)$ be a regular point on $S = \{(x, y) : g(x, y) = 0\}$. That is, $\nabla g(p) \neq 0$. Let $\gamma\colon (-\varepsilon, \varepsilon) \to S$ be a parameterized curve on S with $\gamma(0) = p$. Then $g(\gamma(t)) = 0$ for all t, hence $\frac{d}{dt}g(\gamma(t)) = Dg(\gamma(t))\gamma'(t) = \nabla g(\gamma(t))^T\gamma'(t) = 0$. In particular, $\nabla g(p)$ is orthogonal to $\gamma'(0)$.

Now if p is a point of minimizer of f subjected to $g(x) = 0$, then again it is easy to see that $\nabla f(p)$ is orthogonal to $\gamma'(0)$. In particular, $\nabla f(p)$ and $\nabla g(p)$ are parallel to each other. Thus we have the following theorem:

Theorem 5.2.1. *Let $f\colon \mathbb{R}^2 \to \mathbb{R}$ have continuous 1st order partial derivatives. Let p be a point of minimizer or maximizer of $f(x)$ subjected to $g(x) = 0$, Then $\nabla f(p)$ and $\nabla g(p)$ are parallel to each other. In particular, if $\nabla g(p) \neq 0$, then there exists a scalar λ such that $\nabla f(p) + \lambda \nabla g(p) = 0$. Here λ is called the Lagrange multiplier.* □

The above theorem provides a necessary condition for a point to be a local optimizer. These conditions are called the Lagrange conditions, which consist of two equations

$$\nabla f(p) + \lambda \nabla g(p) = 0 \quad \text{and} \quad g(x) = 0. \tag{5.6}$$

For $n = 2$, we can write the Lagrange conditions as follows:

Let $x^* = (x, y)$ be a point at which $f(x, y)$ has maximum or minimum subjected to $g(x, y) = 0$ then there exists $\lambda \in \mathbb{R}$ such that we have

$$\frac{\partial f}{\partial x} = \lambda \frac{\partial g}{\partial x}, \quad \frac{\partial f}{\partial y} = \lambda \frac{\partial g}{\partial y}, \quad \text{and} \quad g(x, y) = 0. \tag{5.7}$$

This is called the Lagrange condition. The solution of the problem is obtained by solving the Lagrange conditions (5.7) for (x, y).

We can generalize this result for function of n variables with m constraints.

Theorem 5.2.2 (Lagrange)**.** *Let p be a point of local optimizer of $f\colon \mathbb{R}^n \to \mathbb{R}$ subjected to $g(x) = 0$ where $g = (g_1, \ldots, g_m)\colon \mathbb{R}^n \to \mathbb{R}^m$ with $m \leq n$. Let f and g have continuous first order partial derivatives and that p is a regular point. Then there exists $\lambda = (\lambda_1, \lambda_2 \ldots, \lambda_m) \in \mathbb{R}^m$ such that*

$$Df(p) + \lambda^T Dg(p) = 0 \tag{5.8}$$

Here λ is called the *Lagrange multiplier vector* and its components are called *Lagrange multipliers.*

For convenience, we introduce so-called Lagrangian function $L\colon \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}$ defined by

$$\ell(x, \lambda) := f(x) + \lambda^T g(x) \tag{5.9}$$

Thus the Lagrange condition for p to be local optimizer can be expressed as $D\ell(p, \lambda) = 0$, for some $\lambda = (\lambda_1, \lambda_2 \ldots, \lambda_m) \in \mathbb{R}^m$. In other words, the necessary condition in Lagrange theorem is equivalent to the first order necessary condition for unconstrained optimization applied to the Lagrangian function ℓ.

It is easy to see that $D\ell(p, \lambda) = 0$ is equivalent to

$$\begin{aligned}
\frac{\partial \ell}{\partial x_1} &= \frac{\partial f}{\partial x_1} + \sum_{i=1}^{m} \lambda_i \frac{\partial g_i}{\partial x_1} = 0 \\
\vdots \quad & \quad \vdots \\
\frac{\partial \ell}{\partial x_n} &= \frac{\partial f}{\partial x_n} + \sum_{i=1}^{m} \lambda_i \frac{\partial g_i}{\partial x_n} = 0 \\
\frac{\partial \ell}{\partial \lambda_i} &= g_i(x) = 0 \quad 1 \leq i \leq m
\end{aligned}$$

Proof. **of theorem 5.2.2.** We need to show that there exists a vector $\lambda \in \mathbb{R}^m$ such that $Df(p) + \lambda^T Dg(p) = 0$. That is,

$$\nabla f(p) = -Dg(p)^T \lambda.$$

In particular, it is enough to show that $\nabla f(p) \in \mathcal{R}(Dg(p)^T) = \mathcal{N}(p)$. Here $\mathcal{R}(Dg(p)^T)$ is the range space of the matrix $Dg(p)^T$ and $\mathcal{N}(p)$ is normal space S at p. We know that $\mathcal{N}(p) = T_pS^{\perp}$. Thus we have to show that $\nabla f(p) \in T_pS^{\perp}$.

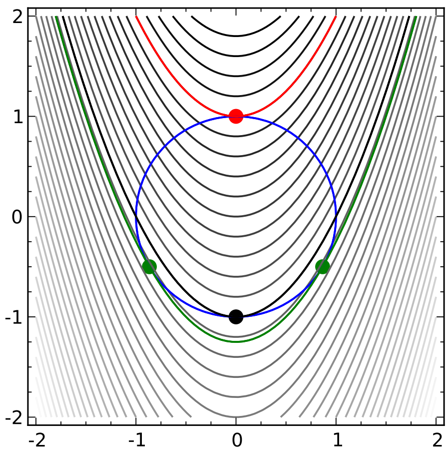

Figure 5.1

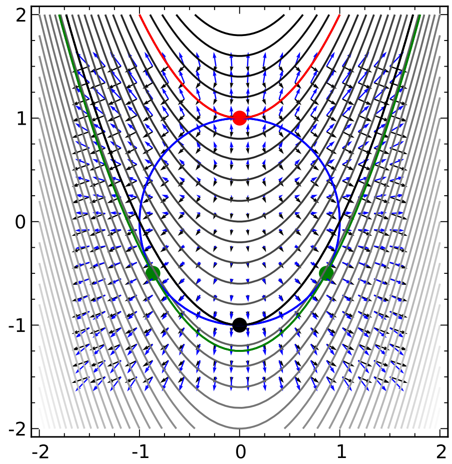

Figure 5.2

For, let $y \in T_pS$. Then there exists a curve $\gamma\colon (-\varepsilon, \varepsilon) \to S$ such that $\gamma(0) = p$ and $\gamma'(0) = y$. Note that $t = 0$ is a local minimizer of $f \circ \gamma$. Therefore,

$$\frac{d}{dt}\left[f \circ \gamma(t)\right]_{t=0} = Df(\gamma(0))\gamma'(0) = Df(p)y = 0.$$

Since $Df(p) = \nabla f(p)^T$, we have $\nabla f(p)^T y = 0$. This completes the proof. $\square$

Example 5.2.1. Optimize $f(x, y) = 4 + x^2 - y$ subjected to $g(x, y) = x^2 + y^2 - 1 = 0$.

Using the Lagrange necessary conditions, there exists λ such that we have the following equations

$$\begin{aligned} 2x &= 2\lambda x \\ -1 &= 2\lambda y \\ x^2 + y^2 &= 1 \end{aligned}$$

From the 1st equation, we have $x(1 - \lambda) = 0$. This mean $x = 0$ or $\lambda = 1$.

If $x = 0$, then $y = \pm 1$. So the points are $(0, \pm 1)$.

If $\lambda = 1$, then from the second equation, we have $y = -1/2$. Hence from 3rd equation, we get $x = \pm\frac{3}{2}$. Thus the points are $(\frac{3}{2}, -\frac{1}{2})$.

We have four possible points of optimizers, $(0, \pm 1)$ and $(\frac{3}{2}, -\frac{1}{2})$. Look at the Figure 5.1. From this figure, it is clear that at these points, the contours of f and g have common tangents. Look at the Figure 5.1, it is clear that at these points the gradients of f and g are parallel to each other.

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Exercise 5.2.2. Show that the Lagrange multipliers do not exist for the following problem minimize $x^2 + y^2$ subjected to $(x - 1)^3 - y^2 = 0$. ‡

5.3 Second order necessary and sufficient conditions

Let us consider the problem:

optimize $f\colon \mathbb{R}^n \to \mathbb{R}$ *subjected to* $g(x) = 0$ *where* $g = (g_1, \ldots, g_m)\colon \mathbb{R}^n \to \mathbb{R}^m, m \leq n.$

Let f and g have continuous second order partial derivatives and $\ell(x,\lambda) := f(x) + \lambda^T g(x)$ is the Lagrangian function. Let $L(x,\lambda), F(x)$ $G_i(x)$ represent Hessian of ℓ, f and g_i respectively. Then

$$L(x,\lambda) = F(x) + \lambda_1 G_1(x) + \lambda_2 G_2(x) + \cdots + \lambda_m G_m(x).$$

Theorem 5.3.1 (Second-Order Necessary Conditions)**.** *Let p be a point of local minimizer of $f\colon \mathbb{R}^n \to \mathbb{R}$ subjected to $g(x) = 0$ where $g = (g_1, \ldots, g_m)\colon \mathbb{R}^n \to \mathbb{R}^m$ and $m \leq n$. Let f and g have continuous second order partial derivatives and that p is a regular point. Then there exists $\lambda = (\lambda_1, \lambda_2 \ldots, \lambda_m) \in \mathbb{R}^m$ such that*
(i) $Df(p) + \lambda^T Dg(p) = 0$
(ii) for all $y \in T_pS$, we have $y^T L(p,\lambda) y \geq 0$.

Remark 5.3.1. Note that $L(p,\lambda)$ plays role similar to the Hessian of unconstrained mimimizer of $\ell(x,\lambda)$. However in this case, we need $y^T L(p,\lambda)y \geq 0$ only on T_pS.

Proof. We have already proved the first condition. To prove the condition (ii), we let $y \in T_pS$. Then there exists a curve $\gamma\colon (-\varepsilon,\varepsilon) \to S$ such that $\gamma(0) = p$ and $\gamma'(0) = y$. Since $t = 0$ is a local minimizer of $f \circ \gamma$, we have

$$\frac{d^2}{dt^2}\left[f \circ \gamma(t)\right]_{t=0} \geq 0.$$

$$\begin{aligned}
\frac{d^2}{dt^2}\left[f \circ \gamma(t)\right]_{t=0} &= \frac{d}{dt}\left[Df(\gamma(t))\gamma'(t)\right]_{t=0} \\
&= \left[\gamma'(t)F(\gamma(t))\gamma'(t) + Df(\gamma(t))\gamma''(t)\right]_{t=0} \\
&= y^T F(p) y + Df(p)\gamma''(0).
\end{aligned}$$

Thus we have

$$y^T F(p) y + Df(p)\gamma''(0) \geq 0. \tag{5.10}$$

Since $\gamma(t) \in S$ for all $t \in (-\varepsilon,\varepsilon)$, for $\lambda = (\lambda_1, \ldots, \lambda_m) \in \mathbb{R}^m$, we have

$$\frac{d^2}{dt^2}\left[\lambda^T g \circ \gamma(t)\right] = 0 \text{ for all } t \in (-\varepsilon,\varepsilon).$$

Expanding the left hand side, we get

$$\begin{aligned}
\frac{d^2}{dt^2}\left[\lambda^T g \circ \gamma(t)\right] &= \frac{d}{dt}\left[\lambda^T \frac{d}{dt}\left[g \circ \gamma(t)\right]\right] \\
&= \frac{d}{dt}\left[\sum \lambda_i \frac{d}{dt}\left[g_i \circ \gamma(t)\right]\right] \\
&= \frac{d}{dt}\left[\sum \lambda_i Dg_i(\gamma(t))\gamma'(t)\right] \\
&= \sum \lambda_i \frac{d}{dt}\left[Dg_i(\gamma(t))\gamma'(t)\right] \\
&= \sum \lambda_i \left[\gamma'(t)^T G_i(\gamma(t))\gamma'(t)) + Dg_i(\gamma(t))\gamma''(t)\right] \\
&= \gamma'(t)^T[\lambda G(\gamma(t))]\gamma'(t) + \lambda^T Dg(\lambda(t))\gamma''(t).
\end{aligned}$$

Substituting $t = 0$ in the above equation, we get

$$y^T[\lambda G(p)]y + \lambda^T Dg(p)\gamma''(0) = 0 \tag{5.11}$$

Adding Equations 5.10 and 5.11, we get

$$y^T F(p) y + Df(p)\gamma''(0) + y^T[\lambda G(p)]y + \lambda^T Dg(p)\gamma''(0) \geq 0.$$

After simplification, we get

$$y^T[F(p)y + \lambda G(p)]y + (Df(p) + \lambda^T Dg(p))\gamma''(0) \geq 0.$$

Note that $(Df(p) + \lambda^T Dg(p) = 0$. Hence

$$y^T[F(p)y + \lambda G(p)]y \geq 0.$$

This proves the result. □

The above conditions are necessary and not sufficient for a point p to be a local minimizer. Consider the following examples.

Example 5.3.2. Find the extrama of $f(x, y) = x$ on the curve $y^2 + x^4 + x^3 = 0$. In this example, the curve has a gradient of 0 at the point $(0, 0)$. The usual equations used in Lagrange multipliers only find the maximum at $(1, 0)$ - they fail to detect the obvious minimum of f at $(0, 0)$ because we must also check for points where $\nabla g \neq 0$. Look at the figure 5.3.2.

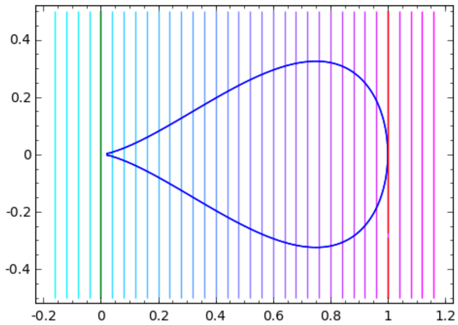

Figure 5.3: Example 5.3.2

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5.4 Second-Order Lagrange Sufficient Conditions

Theorem 5.4.1 (Second-Order Sufficient Conditions)**.** *Let $f\colon \mathbb{R}^n \to \mathbb{R}$ subjected to $g(x) = 0$ where $h = (h_1, \ldots, h_m)\colon \mathbb{R}^n \to \mathbb{R}^m$, Let f and h have continuous second order partial derivatives. If there exist regular point $p \in \mathbb{R}^n$ and $\lambda = (\lambda_1, \lambda_2 \ldots, \lambda_m) \in \mathbb{R}^m$ such that*
(i) $Df(p) + \lambda^T Dh(p) = 0$
(ii) for all $y \in T_pS$, $y \neq 0$, we have $y^T L(p, \lambda)y > 0$
then p is a strict local minimizer of f subjected to $h(x) = 0$.

This theorem states that if p satisfies the first order necessary Lagrangian conditions and that the Hessian $L(p, \lambda)$ is positive definite on T_pS, then p is a strict local minimizer.

Proof. We prove this by contradiction. Suppose p is not a strict local minimum. Then for each $k \in \mathbb{N}$, there exists $y_k \in \Omega$, the feasible set such that $||y_k - p|| < 1/k$ and $f(y_k) \leq f(p)$. Note that in this case the sequence $y_k \to p$. Furthermore, we can write $y_k = p + \delta_k s_k$, with $||s_k|| = 1$ and $0 < \delta_k < 1/k$. Since $s_k \in S^{n-1} := \{x \in \mathbb{R}^n : ||x|| = 1\}$, a compact subset of $\mathbb{R}^n$, the sequence (s_k) has a convergent subsequence, say (s_{k_r}). For simplification of notation or convenience, we assume that the sequence (s_k) itself is convergent and $s_k \to s$.

Note that $h(y_k) - h(p) = 0$. In particular, $\frac{h(y_k)-h(p)}{\delta_k} = 0$, for each k. Also $\delta_k \to 0$ as $k \to \infty$. (Why?), we have

$$\lim_{k\to\infty} \frac{h(p+\delta_k s_k) - h(p)}{\delta_k} = \nabla h(p) \cdot s = 0.$$

Hence $s \in T_p\Omega$. By the Taylor's theorem, for each j, we have

$$0 = h_j(y_k) = h_j(p) + \delta_k \nabla h_j(p) \cdot s_k + \frac{\delta_k^2}{2} s_k^T H_j(\eta_j) s_k$$

for some η_j lying on the line segment joining p and y_k. Multiplying the above equation by λ_j and adding over j, we get

$$\delta_k \sum_j \lambda_j \nabla h_j(p) \cdot s_k + \frac{\delta_k^2}{2} \sum_j \lambda_j s_k^T H_j(\eta_j) s_k = 0 \tag{5.12}$$

Applying, Taylor's theorem to $f(y_k)$ about p, we get

$$0 \geq f(y_k) - f(p) = \delta_k \nabla f(p) \cdot s_k + \frac{\delta_k^2}{2} s_k^T F(\eta_0) s_k \tag{5.13}$$

for some η_0 lying on the line segment joining p and y_k. Adding Eqn. (5.12) and Eqn. (5.13), and combining, we get

$$\delta_k \left[\nabla f(p) + \sum_j \lambda_j \nabla h_j(p) \right] \cdot s_k + \frac{\delta_k^2}{2} s_k^T \left[F(\eta_0) + \sum_j \lambda_j H_j(\eta_j) \right] s_k \leq 0.$$

Note that the first term in the above expression is 0 by condition (i) and $\delta_k \neq 0$, we have

$$s_k^T \left[F(\eta_0) + \sum_j \lambda_j H_j(\eta_j) \right] s_k \leq 0.$$

Note that as $k \to \infty, \eta_k, \eta_0 \to p$, and $s_k \to s \in T_p(S)$. This implies

$$s^T \left[F(p) + \sum_j \lambda_j H_j(p) \right] s \leq 0,$$

which is contradiction to the condition (3). □

Exercise 5.4.1. State the second order sufficient condition for local maximizer. ‡

5.5 Economic interpretation of Lagrange multipliers

A useful aspect of the Lagrange multiplier method is that the values of the multipliers at solution points often has some significance. Mathematically, a multiplier λ_i is the value of the partial derivative of $\ell(x, \lambda)$ with respect to the constraint g_i. So it is the rate at which we could increase the Lagrangian if we were to raise the target of that constraint (from zero). Note that at solution points (p, λ), $\ell(p, \lambda) = f(p)$. Therefore, the rate of increase of the Lagrangian with respect to that constraint is also the rate of increase of the maximum constrained value of with respect to that constraint.

In economics, when f is a profit function and g_i the are constraints on resource amounts, λ_i would be the amount (possibly negative!) by which profit would rise if one were allowed one more unit of resource i . This rate is called the *shadow price* of i

Let us dwell, more into this interpretation.

Consider a minimization problem

$$\text{Minimize } f(x_1, x_2) \text{ subjected to } h(x_1, x_2) = b$$

Where b represent the availability of a certain scarce resource. The Lagrangian of this is given by

$$\ell(x; \lambda) = \ell(x_1, x_2; \lambda) := f(x_1, x_2) - \lambda(h(x_1, x_2) - b).$$

Suppose λ^0 is the value of the Lagrange multiplier corresponding to the minimizer $x^{(0)} = (x_1^{(0)}, x_2^{(0)})$. That is, $h(x^{(0)}) = b$ and

$$\ell(x^{(0)}; \lambda^0) = f(x^{(0)}) = f^0.$$

Clearly f^0 is a function of b, the availability of the resources.

The change ins the optimal value f^0, due to change in b is given by the partial derivative

$$\frac{\partial f^0}{\partial b} = \frac{\partial f^0}{\partial x_1^{(0)}} \frac{\partial x_1^{(0)}}{\partial b} + \frac{\partial f^0}{\partial x_2^{(0)}} \frac{\partial x_2^{(0)}}{\partial b}. \tag{5.14}$$

The partial derivative of the constraint $h(x_1, x_2) - b = 0$ is given by

$$\frac{\partial h}{\partial x_1^{(0)}} \frac{\partial x_1^{(0)}}{\partial b} + \frac{\partial h}{\partial x_2^{(0)}} \frac{\partial x_2^{(0)}}{\partial b} - 1 = 0. \tag{5.15}$$

Multiplying Eq. (5.15) by λ_0 and subtracting from the Eq. (5.14), we get

$$\frac{\partial f^0}{\partial b} = \lambda^0 + \sum_{i=1}^{2} \left[\frac{\partial f^0}{\partial x_i^{(0)}} - \lambda^0 \frac{\partial h}{\partial x_i^{(0)}} \right] \frac{\partial x_i^{(0)}}{\partial b} = \lambda^0. \tag{5.16}$$

Note that the term is the bracket is zero, due to the 1st order necessary Lagrangian conditions.

Thus the rate of change of the optimal value of f with respect to the available resources b is given by the optimal value of the Lagrange multiplier. In other words, the change in the optimal value of the objective function per unit increase in right hand side of the constraint, that is, b, which is defined as shadow price is given by the Lagrange multiplier. Depending upon sign of λ_0, f^0 may increase or decrease.

Example 5.5.1. A household chooses quantities of two food items, *Rice* (quantity x_1) and *Lentils* (quantity x_2). The household's preferences are represented by a Cobb–Douglas utility function

$$U(x_1, x_2) = \sqrt{x_1 x_2} = x_1^{1/2} x_2^{1/2}.$$

Prices and nutritional content are given as follows:

$$\begin{aligned} &p_1 = 2 \quad \text{(price per unit of Rice)}, \quad p_2 = 1 \quad \text{(price per unit of Lentils)}, \\ &c_1 = 50 \quad \text{(calories per unit of Rice)}, \quad c_2 = 20 \quad \text{(calories per unit of Lentils)}. \end{aligned}$$

The household has income (budget) $M = 100$ and must satisfy a calorie requirement $C = 2300$.

We assume both constraints bind and so formulate the equality-constrained maximization problem:

$$\begin{aligned} &\max_{x_1, x_2 \geq 0} \ U(x_1, x_2) = \sqrt{x_1 x_2}, \\ &\text{subject to} \quad 2x_1 + x_2 = 100, \qquad 50x_1 + 20x_2 = 2300. \end{aligned} \tag{5.17}$$

Introduce Lagrange multipliers λ (for the budget) and μ (for the calorie requirement) and form the Lagrangian

$$\ell(x_1, x_2, \lambda, \mu) = \sqrt{x_1 x_2} + \lambda\,(100 - 2x_1 - x_2) + \mu\,(2300 - 50x_1 - 20x_2).$$

The first-order necessary conditions are:

$$\frac{\partial \ell}{\partial x_1} = \tfrac{1}{2}x_1^{-1/2}x_2^{1/2} - 2\lambda - 50\mu = 0, \tag{5.18}$$

$$\frac{\partial \ell}{\partial x_2} = \tfrac{1}{2}x_2^{-1/2}x_1^{1/2} - \lambda - 20\mu = 0, \tag{5.19}$$

$$\frac{\partial \ell}{\partial \lambda} = 100 - 2x_1 - x_2 = 0, \tag{5.20}$$

$$\frac{\partial \ell}{\partial \mu} = 2300 - 50x_1 - 20x_2 = 0. \tag{5.21}$$

Solve the constraints for the feasible bundle

Equations (5.20)–(5.21) are two linear equalities in x_1, x_2:

$$2x_1 + x_2 = 100,$$
$$50x_1 + 20x_2 = 2300.$$

Solve this linear system. Multiply the first equation by 20 to eliminate x_2 when subtracting:

$$40x_1 + 20x_2 = 2000.$$

Subtracting from the second constraint gives

$$(50x_1 + 20x_2) - (40x_1 + 20x_2) = 2300 - 2000 \implies 10x_1 = 300 \implies x_1^* = 30.$$

Substitute back: $2 \cdot 30 + x_2 = 100$ so $x_2^* = 40$.

Hence the optimal (feasible) consumption bundle is

$$x_1^* = 30, \qquad x_2^* = 40.$$

The realized utility is

$$U^* = \sqrt{30 \cdot 40} = \sqrt{1200} = 20\sqrt{3}.$$

Evaluate the left-hand sides of (5.18)–(5.19) at x_1^*, x_2^*. Note

$$\sqrt{\frac{x_2^*}{x_1^*}} = \sqrt{\frac{40}{30}} = \sqrt{\frac{4}{3}} = \frac{2}{\sqrt{3}}, \qquad \sqrt{\frac{x_1^*}{x_2^*}} = \sqrt{\frac{30}{40}} = \sqrt{\frac{3}{4}} = \frac{\sqrt{3}}{2}.$$

Thus the FOCs become the linear system for λ, μ:

$$\tfrac{1}{2}\sqrt{\frac{x_2^*}{x_1^*}} = 2\lambda + 50\mu \quad \implies \quad \frac{1}{\sqrt{3}} = 2\lambda + 50\mu, \tag{5.22}$$

$$\tfrac{1}{2}\sqrt{\frac{x_1^*}{x_2^*}} = \lambda + 20\mu \quad \implies \quad \frac{\sqrt{3}}{4} = \lambda + 20\mu. \tag{5.23}$$

Solve (5.22)–(5.23). From (5.22):

$$2\lambda = \frac{1}{\sqrt{3}} - 50\mu \implies \lambda = \frac{1}{2\sqrt{3}} - 25\mu.$$

Substitute into (5.23):

$$\frac{\sqrt{3}}{4} = \frac{1}{2\sqrt{3}} - 25\mu + 20\mu = \frac{1}{2\sqrt{3}} - 5\mu.$$

Rearrange to solve for μ:

$$-5\mu = \frac{\sqrt{3}}{4} - \frac{1}{2\sqrt{3}} = \sqrt{3}\Big(\frac{1}{4} - \frac{1}{6}\Big) = \sqrt{3}\Big(\frac{3-2}{12}\Big) = \frac{\sqrt{3}}{12},$$
$$\implies \mu^* = -\frac{\sqrt{3}}{60}.$$

Now compute λ^*:

$$\lambda^* = \frac{1}{2\sqrt{3}} - 25\mu^* = \frac{1}{2\sqrt{3}} - 25\Big(-\frac{\sqrt{3}}{60}\Big) = \frac{\sqrt{3}}{6} + \frac{25\sqrt{3}}{60} = \frac{10\sqrt{3}+25\sqrt{3}}{60} = \frac{35\sqrt{3}}{60} = \frac{7\sqrt{3}}{12}.$$

So the exact multipliers are

$$\boxed{\lambda^* = \frac{7\sqrt{3}}{12} \approx 1.01036297, \qquad \mu^* = -\frac{\sqrt{3}}{60} \approx -0.02886751.}$$

Economic (shadow-price) interpretation

By the envelope theorem (or standard sensitivity results for equality-constrained optimization), the Lagrange multipliers equal the marginal change in the optimal objective when the corresponding right-hand side is perturbed, holding other data fixed. Concretely:

$$\frac{\partial U^*}{\partial M} = \lambda^*, \qquad \frac{\partial U^*}{\partial C} = \mu^*.$$

Thus:

- $\lambda^* = 7\sqrt{3}/12$ is the *marginal utility of income*. One additional unit of income raises the maximum attainable utility by exactly $7\sqrt{3}/12$ utility-units (about 1.01036).
- $\mu^* = -\sqrt{3}/60$ is the *marginal effect of the calorie requirement*. Increasing the calorie requirement by one unit (i.e. tightening the constraint) reduces the maximum attainable utility by exactly $\sqrt{3}/60$ utility-units (about 0.02887). Conversely, relaxing the calorie requirement by one unit raises utility by the same magnitude.

$\diamondsuit$

Example 5.5.2. Minimize $f = 9-8x_1-6x_2-4x_3+2x_1^2+2x_2^2+x_3^2+2x_1x_2+2x_1x_3$ subject to $x_1+x_2+2x_3 = 3$.
Solving this problem using the Lagrange multipliers gives

$$x_1^* = \frac{4}{3}, x_2^* = \frac{7}{9}, x_3 = \frac{4}{9}, \lambda^* = \frac{2}{9} \text{ and } f^* = \frac{1}{9}.$$

Find the effect on f^* of changing the constraint to (a) $x_1 + x_2 + 2x_3 = 4$ and (b) $x_1 + x_2 + 2x_3 = 2$.
Note that in the given problem we have $b = 3$ which has increased by 1, that is $db = 1$. Since $\frac{\partial f^*}{\partial b} = \lambda^*$, we have $df = \lambda^* \times db = \frac{2}{9} \times 1 = \frac{2}{9}$. Hence $f^*_{new} = f^* + df = \frac{1}{9} + \frac{2}{9} = \frac{1}{3}$.
Similarly in case of b $f^*_{new} = \frac{1}{9} - \frac{2}{9} = -\frac{1}{9}$. $\diamondsuit$

Example 5.5.3. Optimize $4x_1^2 + x_2^2$ subjected to $2x_1^2 + x_2^2 = 1$.
Let $S = \{(x_1, x_2) \in \mathbb{R}^2 : g(x_1, x_2 = 1 - 2x_1^2 - x_2^2 = 0\}$.
Define $\ell(x_1, x_2, \lambda) = 4x_1^2 + x_2^2 + \lambda(1 - 2x_1^2 - x_2^2)$.
Then using the 1st order necessary condition for (x, y) to a local optimizer, we

$$8x_1 - 4\lambda x_1 = 0 \tag{5.24}$$
$$2x_1 - 2\lambda x_1 = 0 \tag{5.25}$$
$$2x_1^2 + x_2^2 = 1 \tag{5.26}$$

Solving the above three equations we get point $(0, \pm 1, 1)$ and $(\pm 1/\sqrt{2}, 0, 2)$, Thus $(0, \pm 1)$ and $(\pm 1/\sqrt{2}, 0)$ are points of extemizers.

Let us consider $p = (1/\sqrt{2}, 0)$.

$$\begin{aligned} T_pS &= \{(y_1, y_2) \in \mathbb{R}^2 : Dg(p)y = 0\} \\ &= \{(y_1, y_2) \in \mathbb{R}^2 : [2 \times 1/\sqrt{2}, 0][y_1, y_2]^T = 0\} = \{(0, \alpha) : \alpha \in \mathbb{R}\} \end{aligned}$$

The Hessian at $(p, \lambda) = (1/\sqrt{2}, 0, 2)$ is given by

$$L(1/\sqrt{2}, 0, 2) = \begin{pmatrix} 0 & 0 \\ 0 & -2 \end{pmatrix}$$

Let $0 \neq y = (0, \alpha) \in T_pS$, then

$$y^T L(p, \lambda) y = [0, \alpha] \begin{pmatrix} 0 & 0 \\ 0 & -2 \end{pmatrix} \begin{pmatrix} 0 \\ \alpha \end{pmatrix} = -2\alpha^2 < 0$$

Hence $p = (1/\sqrt{2}, 0)$ is a strict local maximizer. Similarly $(-1/\sqrt{2}, 0)$ is a point of strict local maximizer

It is easy to $(0, \pm 1)$ are points of strict local minimizer. See Fig.5.4 for visualization.

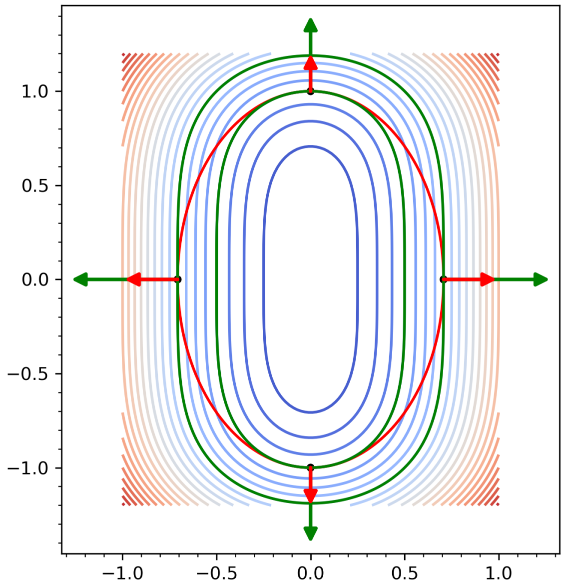

Figure 5.4: Congtours and gradents for Eg.5.5.3

$\diamond$

Example 5.5.4. Find the dimensions of a cylindrical Tin (with bottom and top) made up of a metal sheet to maximize its volume such that the total surface area is 24π.

Let x_1 and x_2 be the radius of the base and height of the cylinder. Then the total surface area of the metal sheet is $2\pi x_1^2 + 2\pi x_1 x_2$ and the volume is $\pi x_1^2 x_2$. Thus mathematically we can write the above problem as:

$$\text{maximize } \pi x_1^2 x_2 \text{ subjected to } 2\pi x_1^2 + 2\pi x_1 x_2 = 24\pi.$$

Using the Lagrange conditions we get

$$\begin{aligned}
2\pi x_1 x_2 &= \lambda(4\pi x_1 + 2\pi x_2) \\
\pi x_1^2 &= 2\pi\lambda x_1 \\
2\pi x_1^2 + 2\pi x_1 x_2 &= 24\pi.
\end{aligned}$$

After solving we get $x_1 = \frac{1}{2}x_2$. Suppose $(x_1^*, x_2^*, \lambda^*)$ is a solution of the above system of equation, then further using the constrained we obtain the solution

$$x_1^* = 2, x_2^* = 4, \lambda^* = -1, f = 16\pi.$$

$\diamondsuit$

Example 5.5.5. The cone $z^2 = x^2 + y^2$ is cut by the plane $z = 1 + x + y$ in some curve C. Find the point on C that is closest to the origin.

This problem can be stated as

$$\text{minimize } f = x_1^2 + x_2^2 + x_3^2 \text{ subjected to } h_1 = x_3^2 - x_1^2 - x_2^2 = 0, h_2 = x_3 - 1 - x_1 - x_2 = 0.$$

We define the Lagrangian function

$$\begin{aligned}
\ell(x_1, x_2, x_3, \lambda_1, \lambda_2) &= f - \lambda_1 h_1 - \lambda_2 h_2 \\
&= x_1^2 + x_2^2 + x_3^2 - \lambda_1(x_3^2 - x_1^2 - x_2^2) - \lambda_2(x_3 - 1 - x_1 - x_2)
\end{aligned}$$

Using the necessary condition for optimization we get the following equations:

$$\begin{aligned}
2x_1 + 2\lambda_1 x_1 + \lambda_2 &= 0 && (5.27) \\
2x_2 + 2\lambda_1 x_2 + \lambda_2 &= 0 && (5.28) \\
2x_3 - 2\lambda_1 x_3 - \lambda_2 &= 0 && (5.29) \\
x_3^2 &= x_1^2 + x_2^2 && (5.30) \\
x_3 &= 1 + x_1 + x_2 && (5.31)
\end{aligned}$$

Now adding (5.29) to (5.27) and (5.28) we reduce the above system to 4 equations in 4 unknowns;

$$\begin{aligned}
x_1 + \lambda_1 x_1 + x_3 - x_3\lambda_1 &= 0 && (5.32) \\
x_2 + \lambda_1 x_2 + x_3 - x_3\lambda_1 &= 0 && (5.33)
\end{aligned}$$

Now we need to solve equations (5.30)-(5.33) for x_1, x_2, x_3, λ_1.
From (5.32) we get $\lambda_1 = \dfrac{x_3 + x_1}{x_3 - x_1}$ and from (5.33) we get $\lambda_1 = \dfrac{x_3 + x_2}{x_3 - x_2}$.
Equating the two values of λ_1 and simplifying we get

$$\frac{x_3 + x_1}{x_3 - x_1} = \frac{x_3 + x_2}{x_3 - x_2} \Longrightarrow x_3(x_1 - x_2) = 0$$

This implies either $x_3 = 0$ or $x_1 = x_2$.

Case I If $x_3 = 0$ then from (5.32), we have

$$x_1^2 + x_2^2 = 0$$

and (5.33), we get

$$1 + x_1 + x_2 = 0.$$

Using the the above tow equations and solving for x_1 we get $2x_1^2 + 2x_1 + 1 = 0$ which does not have a real solution. Thus $x_3 = 0$ is not possible.

Case II if $x_1 = x_2$.
In this case from (5.32), we have
$$x_3^2 = x_1^2 + x_2^2 = 2x_1^2$$
and (5.33), we get
$$x_3 = 1 + x_1 + x_2 = 1 + 2x_2.$$
Thus
$$2x_1^2 = x_3^2 = (1 + 2x_1)^2 \implies 2x_1^2 + 4x_1 + 1 = 0$$
Solving this equation for x_1 we get $x_1 = -1 \pm \frac{1}{\sqrt{2}}$. Now
$$x_3 = 1 + 2x_1 = -1 \pm \sqrt{2}.$$
Thus the critical points (possible optimizers) are
$$p_1 = (-1 + \frac{1}{\sqrt{2}}, -1 + \frac{1}{\sqrt{2}}, -1 + \sqrt{2}) \text{ and } p_2 = (-1 - \frac{1}{\sqrt{2}}, -1 - \frac{1}{\sqrt{2}}, -1 - \sqrt{2}).$$

It is easy to see that for $p_1, \lambda_1 = 2\sqrt{2} - 3, \lambda_2 = 8\sqrt{2} - 12$ and for $p_2, \lambda_1 = -2\sqrt{2} - 3, \lambda_2 = -8\sqrt{2} - 12$
Note that
$$f(p_1) = f(-1 + \frac{1}{\sqrt{2}}, -1 + \frac{1}{\sqrt{2}}, -1 + \sqrt{2}) = 6 - 4\sqrt{(2)}$$
and
$$f(p_2) = f(-1 - \frac{1}{\sqrt{2}}, -1 - \frac{1}{\sqrt{2}}, -1 - \sqrt{2}) = 6 + 4\sqrt{(2)}.$$
Hence $p_2 = (-1 - \frac{1}{\sqrt{2}}, -1 - \frac{1}{\sqrt{2}}, -1 - \sqrt{2})$ is a point of local maximum and $p_1 = (-1 + \frac{1}{\sqrt{2}}, -1 + \frac{1}{\sqrt{2}}, -1 + \sqrt{2})$ is a point of local minimum.

Let us explore the second order conditions. It us easy to see that at p_1
$$T_{p_1}S = \{(t, -t, 0) : t \in \mathbb{R}\}.$$
Similarly
$$T_{p_2}S = \{(t, -t, 0) : t \in \mathbb{R}\}.$$

At p_1, the Hessian of the Lagrangian is
$$L(p_1; \lambda_1, \lambda_2) = \begin{pmatrix} 4\sqrt{2} - 4 & 0 & 0 \\ 0 & 4\sqrt{2} - 4 & 0 \\ 0 & 0 & -4\sqrt{2} + 8 \end{pmatrix}.$$
For $v \in T_{p_1}$,
$$v^T L(p_1; \lambda_1, \lambda_2) v = 8t^2(\sqrt{2} - 1) > 0.$$
Hence p_1 is point of strict local minimizer.

Similarly at p_2,
$$L(p_1; \lambda_1, \lambda_2) = \begin{pmatrix} -4\sqrt{2} - 4 & 0 & 0 \\ 0 & -4\sqrt{2} - 4 & 0 \\ 0 & 0 & 4\sqrt{2} + 8 \end{pmatrix}.$$
For $v \in T_{p_2}$,
$$v^T L(p_2; \lambda_1, \lambda_2) v = -8t^2(\sqrt{2} + 1) < 0.$$
Hence p_2 is point of strict local maximizer. ♢

Exercise 5.5.6. The volume of sales f of a product is found to be a function of the number of social media advertisement x_1 and the number of minutes of television time y as $f(x, y) = 12xy - 3y^2$. Each newspaper social media or each minute on television costs \$1000. How should the firm allocate \$48,000 between the two advertising media for maximizing its sales? ‡

Example 5.5.7. Let P and Q be positive definite matrices of order $n \times n$. Solve the problem: min $x^t Q x$ subjected to $x^t P x = 1$ for any $x \in \mathbb{R}^n$.

It can be shown that if the solution exists then it is an eigenvalues of $P^{-1}Q$.

The Lagrangian function is given by

$$\ell(x, \lambda) = x^t Q x + \lambda(1 - x^t P x).$$

The necessary conditions for ℓ to have extremum at (x^*, λ^*) are

$$\frac{\partial \ell}{\partial x}\ell(x, \lambda)\bigg|_{x=x^*, \lambda=\lambda^*} = 2(x^*)^t Q - 2\lambda^* x^* P = 0, \quad (x^*)^t P x = 1$$

Since P is non-singular, multiplying the 1st equation by P^{-1} both sides, we get

$$P^{-1}Qx^* = \lambda^* x^*.$$

This is the equation for the eigenvalues λs of the matrix $P^{-1}Q$. ♢

Example 5.5.8. A department store plans a one-story rectangular building with required floor area

$$A = 22\,500 \text{ ft}^2$$

and height

$$H = 18 \text{ ft}.$$

Three sides of the building have brick walls and the fourth side is a glass wall. Let x and y (ft) denote the planform side lengths, and assume the glass wall is the full side of length x. Thus the floor area constraint is

$$xy = 22\,500, \qquad x > 0,\ y > 0.$$

Let the unit cost per unit area of brick be $c > 0$. Then unit costs are:

$$\text{brick: } c, \qquad \text{glass: } 2c, \qquad \text{roof: } 3c.$$

Wall and roof areas:

$$\text{brick walls area} = (x + 2y)H, \qquad \text{glass wall area} = xH, \qquad \text{roof area} = xy.$$

Total cost (drop overall positive factor c when minimizing; set $c = 1$ for relative cost):

$$\begin{aligned} C(x, y) &= 1 \cdot (x + 2y)H + 2 \cdot xH + 3 \cdot xy \\ &= H(3x + 2y) + 3xy. \end{aligned}$$

With $H = 18$ ft:

$$C(x, y) = 54x + 36y + 3xy.$$

♢

Example 5.5.9. A funnel in the form of a right circular cone (open at the top) has base radius r, height h, and slant height $l = \sqrt{r^2 + h^2}$. The volume is specified as

$$V = \tfrac{1}{3}\pi r^2 h = 200 \text{ in}^3.$$

The lateral surface area to be minimized is

$$A_{\text{lat}}(r, h) = \pi r l = \pi r \sqrt{r^2 + h^2}.$$

Formulation

Minimize $A_{\mathrm{lat}}(r, h)$ subject to the constraint $\frac{1}{3}\pi r^2 h = 200$, with $r > 0,\ h > 0$.

Eliminate h via the volume constraint:

$$h = \frac{3V}{\pi r^2} = \frac{600}{\pi r^2} \equiv \frac{k}{r^2}, \qquad k = \frac{600}{\pi}.$$

Thus the single-variable objective is

$$A(r) = \pi r\sqrt{r^2 + \frac{k^2}{r^4}}, \qquad r > 0.$$

To simplify differentiation, minimize $\Phi(r) = A(r)^2 = \pi^2\left(r^4 + \frac{k^2}{r^2}\right)$.

◊

Example 5.5.10 (Economics). Consider a consumer whose utility function is

$$U(C_1, C_2, C_3) = 2\sqrt{C_1 C_2 C_3}$$

with an income of Y USD where the price of good 1 is P_1, the price of good 2 is P_2 and the price of good 3 is P_3.

The problem is to max the utility $U(C_1, C_2, U_3)$, subjected to the constrained

$$P_1 C_1 + P_2 C_2 + P_3 C_3 = Y.$$

◊

Example 5.5.11 (Geometry). What shape should a rectangular box with a specific volume (in three dimensions) be in order to minimize its surface area? (Questions like this are very important for businesses that want to save money on packing materials.) Some people may be able to guess the answer intuitively, but we can prove it using Lagrange multipliers. ◊

Example 5.5.12 (Physics). An important application of Lagrange multipliers to Physics:– the Boltzman distribution of Statistical Mechanics. Consider a system of N independent, identical and distinguishable atoms or molecules, for example, a container of oxygen gas. By "distinguishable" we mean that we can index each molecule uniquely and keep track of it. We assume that each molecule can exist in one of n states, $1, 2, \ldots, n$, with respective energies $E_1, E_2, \ldots, E_n$. We will also let N_k denote the "occupation number" of the k-th state (the number of molecules in that state). We assume two conditions:

$$N_1 + N_2 + \cdots + N_n = N.$$

$$N_1 E_1 + N_2 E_2 + \cdots + N_n E_n = E.$$

How many different ways are there of arranging N different molecules among these different states?

This is a combinatorial problem with which you may be familiar. The answer is

$$W(N_1, \ldots, N_n) = \frac{N!}{N_1! N_2! \cdots N_n!}.$$

◊

Determination of Chemical Equilibrium: The problem on chemical equilibrium was proposed by Bracken and McCormic in 1968. This problem is to define the mixture composition of different chemicals when the mixture is at the state of chemical equilibrium. According to the second law of thermodynamics, the free energy of a mixture of chemicals reaches its minimum value at a constant temperature and pressure when the mixture is in chemical equilibrium condition. Therefore, by minimizing the free energy of the mixture, we can determine the chemical composition of any mixture satisfying the state of chemical equilibrium. For describing this system, we will consider the following notations as given below

m :	number of chemical elements in the mixture
n :	number of compounds in the mixture
x_j :	number of moles for compound j, $j = 1, \ldots n$
s :	total number of moles in mixture, $s = \sum x_j$
a_{ij} :	number of atoms of element i in a molecule of compound j
b_i :	atomic weight of element i in the mixture $i = 1, \ldots, n$

The constraint equations for the mixture are given below.

All compounds should have a non negative number of moles, that is,

$$x_j \geq 0, \quad \text{for } j = 1, \ldots n.$$

There is a material balance equation for each element. These equations are represented by linear equality constraint.

$$\sum_{j=1}^{n} a_{ij} x_j = b_i, \quad i = 1, \ldots, m.$$

Here, the total free energy of the mixture is the objective function

$$f(x) = \sum_{j=1}^{n} x_j \left[c_j + \ln\left(\frac{x_j}{s}\right)\right]$$

where

$$c_j = \left(\frac{F^0}{RT}\right)_j + \ln(P)$$

and $\left(\frac{F^0}{RT}\right)_j$ is the model standard free energy function for the j-th compound. P represents the total pressure in atmospheres. Our aim is to find out the parameters x_j that minimize the objective function $f(x)$ subject to the constraints non-negativity of mole number and linear balance.. Therefore, the optimization problem can be written as

$$\begin{aligned} \text{minimize} \qquad & f(x) = \sum_{j=1}^{n} x_j \left[c_j + \ln\left(\frac{x_j}{s}\right)\right] \\ \text{Subject to :} \qquad & c_j = \left(\frac{F^0}{RT}\right)_j + \ln(P) \\ & \sum_{j=1}^{n} a_{ij} x_j = b_i, \quad i = 1, \ldots, m \\ & x_j \geq 0, \quad \text{for } j = 1, \ldots, n. \end{aligned}$$

5.6 Exercises on Lagrange multipliers

1. Determine the dimensions of a rectangular box, open at the top having volume of 32 cubic feets and requiring the least amount of material for its construction.

2. Suppose the cost of manufacturing a particular type of box is such that the base of the box costs three times as much per square foot as the sides and top. Find the dimensions of the box that minimize the cost for a given volume.

3. Find minimizer/Maximizer of in the following:

   (a) $f(x_1, x_2) = x_1 x_2^3$ subjected to $2x_1 + 3x_2 = 4$.

   (b) x_2 subjected to $x_1^3 + x_2^3 - 3x_1 x_2$

   (c) $x_1 x_2 x_3$ subjected tp $\frac{x_1}{a_1} + \frac{x_2}{a_2} + \frac{x_3}{a_3} = 1$ where $a_1, a_2, a_3 > 0$.

   (d) $x_1^2 - 2x_1 + x_2^2 - x_3^2 + 4x_3$ subjected to $x_1 - x_2 + 2x_3 = 2$.

   (e) $f(x, y, z) = x^2 - y^2$ subjected to $x^2 + 2y^2 + 3z^2 = 1$.

4. Let P and Q be positive definite matrices of order $n \times n$. Solve the problem:
   min $x^t Q x$ subjected to $x^t P x = 1$ for any $x \in \mathbb{R}^n$.

   It can be shown that if the solution exists then it is an eigenvalues of $P^{-1}Q$.

5. Minimize $f(x) = \frac{1}{2} ||x||^2$ subjected to $Ax = b$ where A is an $m \times n$ matrix such that AA^t is invertible.

   It can be shown that the minimizer $x^* = A^t(AA^t)^{-1}b$ and $\lambda^* = (AA^t)^{-1}b$.

6. Maximize $f(x, y, z) = \frac{6xyz}{x+2y+2z}$ subjected to $xyz = 16$.

7. A manufacturer's production is modeled by the Cobb-Douglas function

   $$f(x, y) = 100x^{3/4}y^{1/4},$$

   where x represents the units of labor and y represents the units of capital. Each labor unit costs \$200 and each capital unit costs \$250. The total expenses for labor and capital cannot exceed \$50,000. Find the maximum production level.

8. Find the minimum and maximum of $f(x, y) = xy + yz$ on the unit sphere $x^2 + y^2 + z^2 = 1$.

9. A company produces steel boxes at three different plants in amounts x, y and z respectively, producing an annual revenue of $f(x, y, z) = 8xyz^2 - 200(x + y + z)$. The company is to produce 100 units annually. How should the production be distributed to maximize revenue?

10. Let $\alpha_1, \alpha_2, \ldots, \alpha_n$ be positive numbers. Find the maximum value of $f = \alpha_1 x_1 + \alpha_2 x_2 + \cdots + \alpha_n x_n$ subjected to $x_1^2 + \cdots + x_n^2 = 1$.

11. Find the minimum and maximum value of the function $f(x, y, z) = x$ on the curve of intersection of plane $z = x + y$ and the ellipsoid $x^2 + 2y^2 + 2z^2 = 8$.

5.7 Karush-Kuhn-Tucker (KKT) Method

5.7.1 Introduction

In the last chapter, we dealt with minimization problem with equality constraints. In this chapter, we deal with constrained minimization problems involving both equality and inequality constraints. A generic problem of this type can be written as follows:

$$\boxed{\begin{aligned} &\text{minimize } f(x)\\ &\text{subjected to}\\ h_1(x) = 0, &\ldots, h_m(x) = 0\\ g_1(x) \le 0, &\ldots, g_p(x) \le 0 \end{aligned}} \tag{5.34}$$

This is a minimization problem with m equalities constraints and p inequality constraints. Note that any constrained optimization problem can be transformed into this form.

The above problem can be rewritten in compact form as

$$\boxed{\min\ f(x) \text{ subjected to } h(x) = 0 \text{ and } g(x) \le 0} \tag{5.35}$$

where $f\colon \mathbb{R}^n \to \mathbb{R}, h\colon \mathbb{R}^n \to \mathbb{R}^m, g\colon \mathbb{R}^n \to \mathbb{R}^p$ and $m \le n$.

5.7.2 Basic Terminology

Consider the set

$$\Omega := \{x \in \mathbb{R}^n : h(x) = 0, g(x) \le 0\}.$$

Ω is called the *feasible set or constrain set* of the problem (5.35).

Definition 5.7.1. Let $p \in \Omega$. An inequality constraint $g_i \le 0$ is said to be *active* at p if $g_i(p) = 0$. It is *inactive* if $g_i(p) < 0$.

Definition 5.7.2. Let $p \in \Omega$. Let $I(p)$ denote the index set of active inequality constraints at p, that is

$$I(p) := \{j : g_j(p) = 0\}.$$

Definition 5.7.3. A point $p \in \Omega$ is called a *regular point* if $\nabla h_i(p), \nabla g_j(p)$ for $1 \le i \le m$ and $j \in I(p)$ are linearly independent.

Example 5.7.1. Consider the problem
Minimize $2x + y$, subjected to $g_1 = x^2 + y^2 \le 2, g_2 = xy \le \frac{1}{2}, g_3 = x - y \le 1$.
Look at the Figure 5.5.

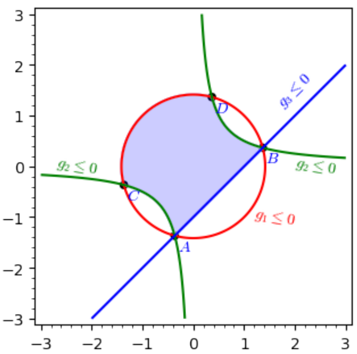

Figure 5.5: Active indices

Note that $I(A) = I(B) = 3$, $I(C) = I(D) = 2$. Suppose P is any point of the circle other than A, B, C, D, then $I(P) = 1$. $\Diamond$

Exercise 5.7.2. Minimize $x + y$, subjected to $g_1 = x^2 + y^2 <= 4, g_2 = 2x + y <= 2, g_3 = y \geq \frac{1}{2}$. Look at the Figure 5.6.

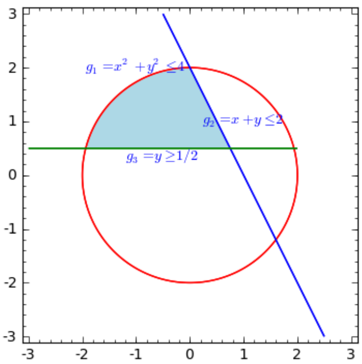

Figure 5.6: Active indices

Find $I(p)$ for various points of the feasible set. $\ddagger$

Feasible and Usable Directions in Constrained Optimization

Consider the constrained minimization problem

$$\min f(x) \quad \text{subject to } g_i(x) \leq 0,\ i = 1, \dots, m, \quad h_j(x) = 0,\ j = 1, \dots, p,$$

where $f, g_i, h_j : \mathbb{R}^n \to \mathbb{R}$ are differentiable. Let

$$\Omega := \{x \in \mathbb{R}^n : h(x) = 0, g(x) \leq 0\}.$$

At a feasible point $x \in \Omega$, a vector $d \in \mathbb{R}^n$ is called a *feasible direction* if for sufficiently small $t > 0$, $x + td \in \Omega$.

This imposes the following first-order conditions:

- For equality constraints $h_j(x) = 0$,

$$h_j(x + td) \approx h_j(x) + t\nabla h_j(x)^T d = 0.$$

  Since $h_j(x) = 0$, we must have

$$\nabla h_j(x)^T d = 0, \quad j = 1, \dots, p.$$

  Thus, d must be tangent to the surface defined by the equality constraints.

- For active inequality constraints $g_i(x) = 0$,

$$g_i(x + td) \approx g_i(x) + t\nabla g_i(x)^T d = t\nabla g_i(x)^T d.$$

  To remain feasible, we require

$$\nabla g_i(x)^T d \leq 0, \quad \text{for all } i \text{ with } g_i(x) = 0.$$

- If $g_i(x) < 0$ (inactive constraint), no restriction on d arises from g_i.

Among feasible directions, we are interested in those that also *decrease the objective function*. For small $t > 0$,

$$f(x + td) \approx f(x) + t\nabla f(x)^T d.$$

If

$$\nabla f(x)^T d < 0,$$

then $f(x + td) < f(x)$, and such a direction d is called a *usable direction* (or *feasible descent direction*).

Geometric Interpretation

- Feasible directions are those that point inside or tangent to the feasible region.
- Usable directions are feasible directions that also point downhill for the objective function.

Example 5.7.3. Consider the optimization problem

$$\min_{x_1,x_2} f(x_1, x_2) = x_1^2 - x_2$$

subject to the constraints

$$\begin{aligned}
g_1(x_1, x_2) &= -x_1 \le 0 && \Rightarrow\ x_1 \ge 0, \\
g_2(x_1, x_2) &= x_1^2 + x_2^2 - 26 \le 0 && \Rightarrow\ x_1^2 + x_2^2 \le 26, \\
g_3(x_1, x_2) &= 6 - x_1 - x_2 \le 0 && \Rightarrow\ x_1 + x_2 \ge 6.
\end{aligned}$$

We want to determine whether each of the directions

$$S_1 = (0, 1), \quad S_2 = (-1, 1), \quad S_3 = (1, 0), \quad S_4 = (-1, 2)$$

is *feasible*, *usable* (descent), or both, at the point

$$p = (5, 1).$$

Step 1: Gradients

The objective gradient is

$$\nabla f(x_1, x_2) = (2x_1,\ -1).$$

At $p = (5, 1)$:

$$\nabla f(p) = (10, -1).$$

Constraint gradients:

$$\nabla g_1(x) = (-1, 0), \quad \nabla g_2(x) = (2x_1, 2x_2), \quad \nabla g_3(x) = (-1, -1).$$

At $p = (5, 1)$:

$$\nabla g_2(p) = (10, 2), \quad \nabla g_3(p) = (-1, -1).$$

Step 2: Active Constraints at $p = (5, 1)$

$$\begin{aligned}
&g_1(p) = -5 < 0 \quad \Rightarrow\ \text{inactive}, \\
&g_2(p) = 25 + 1 - 26 = 0 \quad \Rightarrow\ \text{active}, \\
&g_3(p) = 6 - 5 - 1 = 0 \quad \Rightarrow\ \text{active}.
\end{aligned}$$

So the active set is $\{g_2, g_3\}$.

Step 3: Classification of Directions

Direction $S_1 = (0,1)$:

$$\nabla f(p) \cdot S_1 = (10,-1) \cdot (0,1) = -1 < 0 \quad \Rightarrow \text{ usable.}$$
$$\nabla g_2(p) \cdot S_1 = (10,2) \cdot (0,1) = 2 > 0 \quad \Rightarrow \text{ not feasible.}$$

So S_1 is *usable only*.

Direction $S_2 = (-1,1)$:

$$\nabla f(p) \cdot S_2 = (10,-1) \cdot (-1,1) = -11 < 0 \quad \Rightarrow \text{ usable.}$$
$$\nabla g_2(p) \cdot S_2 = (10,2) \cdot (-1,1) = -8 \leq 0, \qquad \nabla g_3(p) \cdot S_2 = (-1,-1) \cdot (-1,1) = 0 \leq 0.$$

So S_2 is *both feasible and usable*.

Direction $S_3 = (1,0)$:

$$\nabla f(p) \cdot S_3 = (10,-1) \cdot (1,0) = 10 > 0 \quad \Rightarrow \text{ not usable.}$$
$$\nabla g_2(p) \cdot S_3 = (10,2) \cdot (1,0) = 10 > 0 \quad \Rightarrow \text{ not feasible.}$$

So S_3 is *neither feasible nor usable*.

Direction $S_4 = (-1,2)$:

$$\nabla f(p) \cdot S_4 = (10,-1) \cdot (-1,2) = -12 < 0 \quad \Rightarrow \text{ usable.}$$
$$\nabla g_2(p) \cdot S_4 = (10,2) \cdot (-1,2) = -6 \leq 0, \qquad \nabla g_3(p) \cdot S_4 = (-1,-1) \cdot (-1,2) = -1 \leq 0.$$

So S_4 is *both feasible and usable*.

Final Classification

$$\begin{aligned}
S_1 = (0,1): &\qquad \text{Usable only,} \\
S_2 = (-1,1): &\qquad \text{Feasible and usable,} \\
S_3 = (1,0): &\qquad \text{Neither feasible nor usable,} \\
S_4 = (-1,2): &\qquad \text{Feasible and usable.}
\end{aligned}$$

$\Diamond$

5.7.3 Karush-Kuhn-Tucker Conditions

Now we deal with necessary conditions for a regular point to be a solution of the problem (5.35).

Theorem 5.7.1. *Let f, g, h be functions having continuous first order partial derivatives. Let $x^* \in \Omega$ be a regular for the constraints that are active at x^* and a solution of problem* (5.35)*. Then there exist $\lambda^* = (\lambda_1, \ldots, \lambda_m) \in \mathbb{R}^m$ and $\mu^* = (\mu_1, \ldots, \mu_p) \in \mathbb{R}^p$ such that following conditions*

$$\mu^* \geq 0 \tag{5.36}$$
$$Df(x^*) + {\lambda^*}^T Dh(x^*) + {\mu^*}^T Dg(x^*) = 0 \tag{5.37}$$
$${\mu^*}^T g(x^*) = 0 \tag{5.38}$$

are satisfied.

We refer λ^* as *Lagrange multiplier vector* and μ^* as *KKT multiplier vector*. Components of λ^* and μ^* are called *Lagrange multipliers* and *KKT-multipliers* respectively.

The conditions (5.36) are called the *dual feasibility conditions.* The conditions (5.37) are called *stationarity conditions* and conditions (5.38) are called *complementary slackness conditions.*

The condition (5.37), can be written in expanded form as

$$\nabla f(x) + \sum_{i=1}^{m} \lambda_i \nabla h_i(x) + \sum_{j \in I(x^*)} \mu_j \nabla g_j(x).$$

This implies

$$-\nabla f(x) = \sum_{i=1}^{m} \lambda_i \nabla h_i(x) + \sum_{j \in I(x^*)} \mu_j \nabla g_j(x).$$

In particular, at the solution, the negative of the gradient of f can be expressed as linear combination of gradient of constraints. (Refer to the Fig. 5.7.)

Note that since $\mu_j \geq 0$ and $g_j \leq 0$, the condition

$$\mu^{*T} g(x^*) = \sum_{j \in I(x^*)} \mu_j g_j(x^*) = 0,$$

is equivalent to the statement that a component of μ may be nonzero only if the corresponding constraint is active. This is called a *complementary slackness conditions*, stating that

$$g_j(x^*) < 0 \implies \mu_j = 0 \text{ and } \mu_j > 0 \implies g_j(x^*) = 0.$$

The conditions (5.36),(5.37) and (5.38) are called KKT conditions. Any point $p \in \Omega$ which satisfies the *KKT conditions* is called a *KKT point.*

Proof. (of Theorem 5.7.1) Let $\Omega = \{x \in \mathbb{R}^n : h(x) = 0, g(x) \leq 0\}$. Since x^* is a local minimizer of f on Ω, x^* is also a local minimizer on $\Omega' = \{x \in \mathbb{R}^n : h(x) = 0, g_j(x) = 0, j \in I(x)\}$. Hence by the first order necessary conditions of Lagrange multipliers, there exist vectors $\lambda^* \in \mathbb{R}^m$ and μ_j, $j \in I(x^*)$ such that conditions (5.37) and (5.38) are satisfied. Since $\mu_j = 0$ for $j \notin I(x^*)$, conditions (5.37) and (5.38) are satisfied for all $1 \leq i \leq m$ and $1 \leq j \leq p$.

It remains to prove that $\mu_j \geq 0$ for $j \in I(x^*)$. Suppose $\mu_k < 0$ for some $k \in J(x^*)$. Define

$$S = \{x \in \mathbb{R}^n : h(x) = 0, g_j(x) = 0, j \in I(x^*), j \neq k\}.$$

The tangent space at x^* to S is given by

$$T_{x^*}S = \{y \in \mathbb{R}^n : Dh(x^*)y = 0, Dg_j(x^*)y = 0, j \in I(x^*), j \neq k\}.$$

We claim that there exists $y \in T_{x^*}S$ such that $Dg_k(x^*)y \neq 0$. On contrary, let $Dg_k(x^*)y = 0$ for all $y \in T_{x^*}S$, then $Dg_k(x^*)y = \nabla g_k(x^*) \cdot y = 0$. This implies $\nabla g_k(x^*) \in (T_{x^*}S)^\perp$. Hence we have

$$\nabla g_k(x^*) \in \text{Span}\{\nabla h_i(x^*), 1 \leq i \leq m, \nabla g_j(x^*), j \in J(x^*), j \neq k\}.$$

This is a contradiction to the fact that x^* is a regular point.

Without loss of generality, let us assume that $Dg_k(x^*)y < 0$. Post multiplying the condition (5.37) by the vector y, we get

$$Df(x^*)y + \lambda^{*T} Dh(x^*)y + \sum_{j \neq k} \mu_j^* Dg_j(x^*)y + \mu_k^* Dg_k(x^*)y = 0.$$

Since $Dh(x^*)y = 0$ and $Dg_j(x^*)y = 0$ for $j \in J(x^*), j \neq k$, we have,

$$Df(x^*)y = -\mu_k^* Dg_k(x^*)y < 0, \quad \text{since } \mu_k < 0, \text{ and } Dg_k(x^*)y < 0.$$

Since $y \in T_{x^*}S$, there exist a $\mathcal{C}^1$ curve $\gamma\colon (-\varepsilon, \varepsilon) \to S$ such that $\gamma(0) = x^*$ and $\gamma'(0) = y$. Hence

$$\frac{d}{dt}(f \circ \gamma)(t)\bigg|_{t=0} = Df(x^*)y < 0.$$

This implies $f \circ \gamma$ is decreasing in some neighbourhood of 0, that is, there exists, $\delta > 0$ with $\varepsilon > \delta_1 > 0$ such that for all $t \in (-\delta_1, \delta_1)$, $f(\gamma(t)) < f(\gamma(0)) = f(x^*)$.

On the other hand

$$\frac{d}{dt}(g_k \circ \gamma)(t)\bigg|_{t=0} = Dg_k(x^*)y < 0.$$

Hence, there exists $\varepsilon > \delta_2 > 0$ such that for all $t \in (-\delta_2, \delta_2)$, $g_k(\gamma(t)) \le 0$.

Let $\delta = \min\{\delta_1, \delta_2\}$, then for all $t \in (-\delta, \delta)$, we have $f(\gamma(t)) < f(x^*)$ and $g_k(\gamma(t)) \le 0$. Note that $t \in (-\delta, \delta)$, $\gamma(t) \in S$, and we have $f(\gamma(t)) < f(x^*)$. This is contradiction to assumption that x^* is a local minimizer of f.

□

Many times, we are given an optimization problem which is not of the form (5.35). In such cases, we can convert this problem to the standard form and then write the KKT conditions. Let us look at some examples.

Example 5.7.4. Write the KKT condition for the problem

$$\text{maximize} \quad f(x) \text{ subjected to} \quad h(x) = 0 \text{ and } g(x) \le 0 \tag{5.39}$$

The KKT condition can be obtained by replacing f by $-f$ in the KKT conditions (5.36)-(5.38).

1. $\mu^* \ge 0$;
2. $-Df(x^*) + {\lambda^*}^T Dh(x^*) + {\mu^*}^T Dg(x^*) = 0$;
3. ${\mu^*}^T g(x^*) = 0$;
4. $h(x^*) = 0$;
5. $g(x^*) \le 0$;

Equivalently, by changing signs for λ^* and μ^*, we get

1. $\mu^* \le 0$;
2. $Df(x^*) + {\lambda^*}^T Dh(x^*) + {\mu^*}^T Dg(x^*) = 0$;
3. ${\mu^*}^T g(x^*) = 0$;
4. $h(x^*) = 0$;
5. $g(x^*) \le 0$;

◊

Example 5.7.5. The KKT condition for the problem

$$\text{minimize} \quad f(x) \text{ subjected to } h(x) = 0 \text{ and } g(x) \ge 0 \tag{5.40}$$

1. $\mu^* \le 0$;
2. $Df(x^*) + {\lambda^*}^T Dh(x^*) + {\mu^*}^T Dg(x^*) = 0$;
3. ${\mu^*}^T g(x^*) = 0$;
4. $h(x^*) = 0$;
5. $g(x^*) \ge 0$;

◊

Geometric Interpretation of KKT

Look at the Fig. 5.7 for two inequality constraints $g_1, g_2 \leq 0$. In this case, the KKT condition implies $\nabla f(x^*) = -\mu_1 \nabla g_1(x^*) - \mu_1 \nabla g_2(x^*)$ where $\mu_1, \mu_2 > 0$.

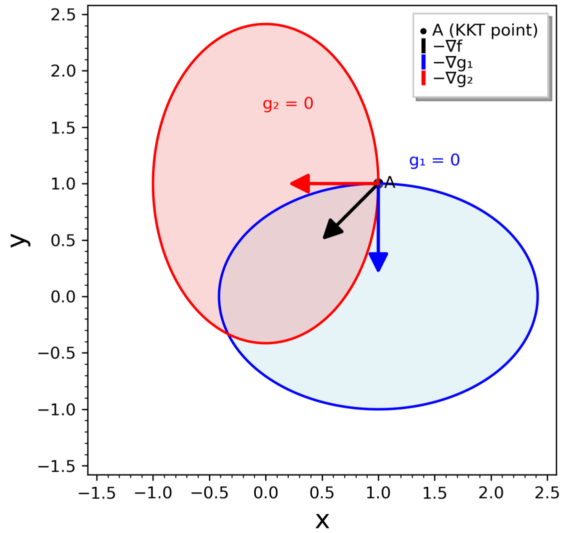

Figure 5.7: Geometric meaning of KKT conditions

Example 5.7.6. Minimize $x_1^2 + x_2^2 - 14x_1 - 6x_2$ subjected to $2 - x_1 - x_2 \geq 0$ and $3 - x_1 - 2x_2 \geq 0$.
We can convert this problem in standard form as
Minimize $x_1^2 + x_2^2 - 14x_1 - 6x_2$ subjected to $x_1 + x_2 - 2 \leq 0$ and $x_1 + 2x_2 - 3 \leq 0$.

Solution: If (x_1, x_2) is a solution of this problem then, there exist $\mu_1, \mu_2 \in \mathbb{R}$ such that the following KKT conditions hold
$\mu_1 \geq 0$, $\mu_2 \geq 0$.
$2x_1 - 14 + \mu_1 + \mu_2 = 0$
$2x_2 - 6 + \mu_1 + 2\mu_2 = 0$
$\mu_1(x_1 + x_2 - 2) = 0$
$\mu_2(x_1 + 2x_2 - 3) = 0$
We need to solve the above equations along with the given constraints for x_1, x_2, μ_1, μ_2. We approach this by taking various cases on indices being active or inactive.

Case I: If no active constraints: $\mu_1^* = \mu_2^* = 0$ implies $x_1^* = 7$ and $x_2^* = 3$. But this violates both constraints. Hence is not a valid point.

Case II: If g_1 is active and g_2 is inactive, then $\mu_2 = 0$. This gives $\mu_1^* = 8$ and $x^* = (3, -1)$. It is easy to check that it is a KKT point and a solution.

Case III: If g_2 is active and g_1 is inactive, then $\mu_1^* = 0$. This implies $\mu_2^* = 4$ and $x^* = (5, -1)$ which is not a KKT point as it violates g_1.

Case IV: Both are active constraints: In this case we get

$$x^* = (1, 1), \mu^* = (20, -8).$$

Since $\mu_2 < 0$ is is not a KKT point.

Thus the only solution is

$$x^* = (3, -1), \mu^* = (8, 0).$$

◊

5.7.4 Second-order Necessary Conditions

We define the *Lagrangian function* related to the optimization problem (5.35) as

$$\ell(x, \lambda, \mu) = f(x) + \lambda^T h(x) + \mu^T g(x).$$

Now define

$$L(x, \lambda, \mu) := F(x) + [\lambda H] + [\mu G]$$

where $F(x)$ is the Hessian of f at x,

$$[\lambda H] = \lambda_1 H_1(x) + \ldots + \lambda_m H_m(x)$$

where H_k is the Hessian of h_k at x and

$$[\mu G] = \mu_1 G_1(x) + \ldots + \mu_p G_p(x)$$

where G_k is the Hessian of g_k at x.

Let S be the surface defined by the equalities and active constraints

$$S = \{x \in \Omega : h(x) = 0, g_j(x) = 0, j \in I(x)\}.$$

The tangent space to the surface S is given by

$$T(x^*) = T_{x^*}S := \{y \in \mathbb{R}^n : Dh(x^*)y = 0, Dg_j(x^*)y = 0, j \in I(x^*)\}.$$

Next, we deal with second order necessary condition for a regular point to be a minimize of f.

Theorem 5.7.2 (Second-order Necessary Conditions)**.** *Let $f\colon \mathbb{R}^n \to \mathbb{R}, g\colon \mathbb{R}^n \to \mathbb{R}^p, h\colon \mathbb{R}^n \to \mathbb{R}^m$ be functions having continuous second partial derivatives. Let $x^* \in \Omega$ be a regular point and a solution of problem* (5.35)*. Then there exist $\lambda^* = (\lambda_1, \ldots, \lambda_m) \in \mathbb{R}^m$ and $\mu^* = (\mu_1, \ldots, \mu_p) \in \mathbb{R}^p$ such that following conditions are satisfied:*

(1) $\mu^* \geq 0$.

(2) $Df(x^*) + {\lambda^*}^T Dh(x^*) + {\mu^*}^T Dg(x^*) = 0$.

(3) ${\mu^*}^T g(x^*) = 0$.

(4) For all $y \in T(x^*)$, $y^T L(x^*, \lambda^*, \mu^*)y \geq 0$.

Proof. Note that conditions (1), (2) and (3) follows from the 1st order necessary conditions of the Theorem (5.7.1). Since x^* is a local minimizer over the feasible set Ω, it is also a local minimizer over the surface $S \subset \Omega$ defined above. Hence using the second order necessary conditions for equality constraints, the condition (4) follows. □

5.7.5 Second-order Sufficient Conditions

Now we deal with second order sufficient condition for a regular point to be a solution of the problem (5.35). We require the sets

$$\hat{I}(x^*, \mu^*) := \{i : g_i(x^*) = 0, \mu_i^* > 0\}.$$

Note that in this case $\hat{I}(x^*, \mu^*) \subset I(x^*)$.

$\hat{T}(x^*, \mu^*) := \{y \in \mathbb{R}^n : Dh(x^*)y = 0, Dg_i(x^*)y = 0, i \in \hat{I}(x^*, \mu^*)\}$.

Since $\hat{I}(x^*, \mu^*) \subset I(x^*)$, $T(x^*) \subset \hat{T}(x^*, \mu)$.

Theorem 5.7.3 (Second-order Sufficient Conditions)**.** *Suppose $f: \mathbb{R}^n \to \mathbb{R}$, $h: \mathbb{R}^n \to \mathbb{R}^m$ and $g: \mathbb{R}^n \to \mathbb{R}^p$ be function having continuous second order partial derivatives. Let x^* be a feasible point and $\lambda^* \in \mathbb{R}^m$ and $\mu^* \in \mathbb{R}^p$ such that*

(1) $\mu^* \geq 0$.

(2) $Df(x^*) + {\lambda^*}^T Dh(x^*) + {\mu^*}^T Dg(x^*) = 0$.

(3) ${\mu^*}^T g(x^*) = 0$.

(4) For all $y \in \hat{T}(x^*, \mu^*)$, $y \neq 0$, we have $y^T L(x^*, \lambda^*, \mu^*)y > 0$.

Then x^* is strict local minimizer of f subjected to $h(x) = 0$ and $g(x) \leq 0$.

Proof. Proof of this theorem is similar to sufficient condition for equality constraints in case of Lagrange multipliers. Assume that x^* is not a strict local minimizer of $f(x)$ subjected to $h(x) = 0$ and $g(x) \leq 0$. Then there exists a sequence $y_k \in \Omega$ such that $||y_k - x^*|| < 1/k$ and $f(y_k) \leq f(x^*)$. We can write y_k as $y_k = x^* + \delta_k s_k$, with $0 < \delta_k < 1/k$ and $||s_k|| = 1$ and $f(y_k) \leq f(x^*)$. Since (s_k) is a sequence in a compact set $S^{n-1} = \{x \in \mathbb{R}^n : ||x|| = 1\}$, it is convergent. Assume that $s_k \to s$. Thus

$$\lim_{k\to\infty} \frac{f(y_k) - f(x^*)}{\delta_k} = \lim_{k\to\infty} \frac{f(x^* + \delta_k s_k) - f(x^*)}{\delta_k} = \nabla f(x^*) \cdot s \leq 0.$$

Similarly for each $i \in \{1, \dots, m\}$, $h_i(y_k) = 0$, we have

$$\lim_{k\to\infty} \frac{h_i(y_k) - h_i(x^*)}{\delta_k} = \nabla h_i(x^*) \cdot s = 0.$$

Also for each active constraint $j \in I(x^*)$, we have $g_j(y_k) - g_j(x^*) \leq 0$ implies $\nabla g_j(x^*) \cdot s \leq 0$.

If $\nabla g_j(x^*) \cdot s = 0$ for each $j \in I(x^*)$, the proof goes through as in case of equality constraints.

If there exists $k \in I(x^*)$ such that $\nabla g_k(x^*) \cdot s < 0$, then

$$0 \geq \nabla f(x^*) \cdot s = -(\lambda^*)^T \nabla h(x^*) \cdot s - (\mu^*)^T \nabla g(x^*) \cdot s > 0.$$

This is a contradiction.

□

Example 5.7.7. Minimize $x_1^2 + x_2^2 - 14x_1 - 6x_2$ subjected to $2 - x_1 - x_2 \geq 0$ and $3 - x_1 - 2x_2 \geq 0$.

We have seen (in the Example 5.7.6) that this problem has only one valid KKT point is

$$x^* = (3, -1), \mu^* = (8, 0).$$

The Lagrangian function is

$$\ell(x, \lambda, \mu) = f(x, y) + \mu_1 g_1(x, y) + \mu_2 g_2(x, y).$$

The Hessian of the Lagrangian is

$$L(x^*, \mu^*) = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} > 0.$$

Note that L positive definite on $\mathbb{R}^n$ and hence x^* is local minimum. $\diamondsuit$

5.7.6 Solved Problems

Example 5.7.8. A manufacturing company producing small refrigerators has entered into a contract to supply 50 refrigerators at end of the next three months. The cost of producing x refrigerators in any month is \$$(x^2 + 1000)$. The firm can produce more refrigerators in any month and carry them to subsequent month. However, it costs \$20 per unit for any refrigerator carried over from one month to the next. Assuming that no initial inventory, determine the number of refrigerators to be produced in each month to minimize the total cost. $\diamondsuit$

Formulation of the problem

Let x_1, x_2, x_3 be number of refrigerators produced in the first, second and third month respectively. The total cost is

$$\begin{aligned} f(x_1, x_2, x_3) &= (x_1^2 + 1000) + (x_2^2 + 1000) + (x_3^2 + 1000) \\ &\quad +20(x_1 - 50) + 20(x_1 + x_2 - 100). \end{aligned}$$

After simplifications we get

$$f(x_1, x_2, x_3) = x_1^2 + x_2^2 + x_3^2 + 40x_1 + 20x_2.$$

The constraints are

$$\begin{aligned} g(x_1, x_2, x_3) &= x_1 - 50 \geq 0. \\ g(x_1, x_2, x_3) &= x_1 + x_2 - 100 \geq 0. \\ g(x_1, x_2, x_3) &= x_1 + x_2 + x_3 - 150 \geq 0. \end{aligned}$$

Solution: Using appropriate KKT conditions, there exist, μ_1, μ_2, μ_3 such that

1. $\mu_1 \leq 0, \mu_1 \leq 0, \mu_1 \leq 0.$
2. $2x_1 + 40 + \mu_1 + \mu_2 + \mu_3 = 0.$
3. $2x_2 + 20 + \mu_2 + \mu_3 = 0.$
4. $2x_3 + \mu_3 = 0.$
5. $\mu_1(x_1 - 50) = 0.$
6. $\mu_2(x_1 + x_2 - 100) = 0.$
7. $\mu_3(x_1 + x_2 + x_3 - 150) = 0.$

We can solve the above equations from 1 to 7 by taking different cases. One can obtain after dealing with quite a few cases, that the only solution to the above problem is

$$x_1 = 50, x_2 = 50, x_3 = 50, \mu_1 = -20, \mu_2 = -20, \mu_3 = -100.$$

Since $L(x^*, \mu^*) = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} > 0$, $x^* = (50, 50, 50)$ on $\mathbb{R}^3$, is a local minimizer.

Example 5.7.9. Minimize $f(x_1, x_2) = (x_1 - 1)^2 + x_2 - 2$, subjected to $h(x_1, x_2) = x_2 - x_1 - 1 = 0$ and $g(x_1, x_2) = x_1 + x_2 - 2 \leq 0$. $\diamondsuit$

First of all note that $\nabla h(x_1, x_2) = [-1, 1]$, $\nabla g(x_1, x_2) = [1, 1]$ which are linearly independent and hence all feasible points are regular.

Lagrangian and KKT conditions are as follows:

$$\begin{gathered} 2x_1 - 2 - \lambda + \mu = 0. \\ 1 + \lambda + \mu = 0. \\ \mu(x_1 + x_2 - 2) = 0. \\ x_2 - x_1 - 1 = 0. \\ x_1 + x_2 - 2 = 0. \end{gathered}$$

Case I: If $\mu > 0$, then $g = 0$ and solving the above equations in KKT conditions gives

$$x^* = (1/2, 3/2), \lambda^* = -1, \mu^* = 0$$

which is not a legitimate solution.

Case II: If $\mu = 0$, Then the solution is given by

$$x^* = (1/2, 3/2), \lambda = -1.$$

This is a KKT point.

$$L(x^*, \lambda^*, \mu^*) = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}.$$

Note that $\mu^* = 0$, the constraint $g(x) = 0$ does not enter in the computation of $\hat{T}(x^*, \mu^*)$.

$$\hat{T}(x^*, \mu^*) = \{(a, a) : a \in \mathbb{R}\}.$$

This means for $0 \neq y \in \hat{T}(x^*, \mu^*)$, we get $y^T L(x^*, \mu^*) y = 2a^2 > 0$. Hence x^* is a local minimum.

Example 5.7.10. Consider the problem

$$\min_{x_1, x_2} f(x_1, x_2) = (x_1 - 1)^2 + (x_2 - 5)^2$$

subject to

$$g_1(x) = -x_1^2 + x_2 - 4 \leq 0, \qquad g_2(x) = -(x_1 - 2)^2 + x_2 - 3 \leq 0.$$

Lagrangian and KKT conditions

$$\ell(x, \lambda) = (x_1 - 1)^2 + (x_2 - 5)^2 + \mu_1(-x_1^2 + x_2 - 4) + \mu_2(-(x_1 - 2)^2 + x_2 - 3).$$

Stationarity:

$$\begin{aligned} \frac{\partial \ell}{\partial x_1} &= 2(x_1 - 1) - 2\mu_1 x_1 - 2\lambda_2(x_1 - 2) = 0, \\ \frac{\partial \ell}{\partial x_2} &= 2(x_2 - 5) + \mu_1 + \lambda_2 = 0. \end{aligned}$$

Complementary slackness:

$$\mu_1 g_1(x) = 0, \qquad \mu_2 g_2(x) = 0,$$

with $\mu_i \geq 0$, $g_i(x) \leq 0$.

Case analysis

Case-I: No active constraints In this case $\mu_1 = \mu_2 = 0$. This leads to a point $(1, 5)$ which violates the the constraint g_2. Hence it is not a valid KKT point.

Case-II: Only g_1 active. This leads to no admissible solution (only $\mu_1 = 0$ arises, contradicting activity).

Case-III: Only g_2 active. From stationarity we obtain $x_2 = 5-\lambda_2/2$ and $x_1 = (1-2\mu_2)/(1-\lambda_2)$. Substituting into $g_2 = 0$ gives

$$(\mu_2 - 2)(\mu_2^2 - 4\mu_2 + 1) = 0,$$

so $\mu_2 \in \{2, 2 \pm \sqrt{3}\}$. The three corresponding points are

$$(3, 4), \quad (\approx 0.6340, \approx 4.8660), \quad (\approx 2.3660, \approx 3.1340).$$

The middle one violates g_1 and is infeasible. The two feasible points give objectives 5 and ≈ 5.348 respectively.

Cas-IV: Both g_1, g_2 active. Solving the two active equalities yields

$$x_1 = \tfrac{3}{4}, \qquad x_2 = \tfrac{73}{16}.$$

Solving stationarity for μ_1, μ_2 gives

$$\mu_1 = \frac{27}{64}, \qquad \mu_2 = \frac{29}{64},$$

both positive. The objective equals

$$f\left(\tfrac{3}{4}, \tfrac{73}{16}\right) = \frac{65}{256} \approx 0.25390625.$$

Second-order condition

$$\nabla^2 f = 2I, \qquad \nabla^2 g_1 = \nabla^2 g_2 = \begin{bmatrix} -2 & 0 \\ 0 & 0 \end{bmatrix}.$$

Thus

$$\nabla_x^2 \ell = 2I - (\mu_1 + \mu_2) \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 2 - 2(\frac{7}{8}) & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} \frac{1}{4} & 0 \\ 0 & 2 \end{bmatrix} \succ 0.$$

So second-order sufficiency holds. By convexity of f, this is the global minimizer.

Final Answer

The global minimizer is

$$x^* = \left(\tfrac{3}{4}, \tfrac{73}{16}\right) = (0.75, 4.5625),$$

with multipliers

$$\lambda_1 = \tfrac{27}{64}, \qquad \lambda_2 = \tfrac{29}{64},$$

and minimum value

$$f(x^*) = \tfrac{65}{256} \approx 0.2539.$$

◊

Example 5.7.11. Minimize
$$f(x) = x_1^2 + x_2^2 + x_3^2$$
subject to
$$x_1 + x_2 + x_3 \geq 5, \qquad 2 - x_2x_3 \leq 0, \qquad x_1 \geq 0,\ x_2 \geq 0,\ x_3 \geq 2.$$
Write constraints as $g_i(x) \leq 0$:
$$\begin{aligned} g_1(x) &:= 5 - (x_1 + x_2 + x_3) \leq 0, \\ g_2(x) &:= 2 - x_2x_3 \leq 0, \\ g_3(x) &:= -x_1 \leq 0, \\ g_4(x) &:= -x_2 \leq 0, \\ g_5(x) &:= 2 - x_3 \leq 0. \end{aligned}$$

Lagrangian with multipliers $\mu_i \geq 0$:
$$\ell(x, \mu) = x_1^2 + x_2^2 + x_3^2 + \sum_{i=1}^{5} \mu_i g_i(x).$$

Stationarity $\nabla_x \ell = 0$ gives
$$\begin{aligned} 2x_1 - \mu_1 - \mu_3 &= 0, \\ 2x_2 - \mu_1 - \mu_2 x_3 - \mu_4 &= 0, \\ 2x_3 - \mu_1 - \mu_2 x_2 - \mu_5 &= 0. \end{aligned}$$

We examine the candidate points
$$p_1 = \left(\tfrac{3}{2}, \tfrac{3}{2}, 2\right), \quad p_2 = \left(\tfrac{4}{3}, \tfrac{2}{4}, 3\right) = (4/3, 0.5, 3), \quad p_3 = (2, 1, 2).$$

Check for $p_1 = (\frac{3}{2}, \frac{3}{2}, 2)$. Evaluate constraints:
$$g_1(p_1) = 0, \quad g_2(p_1) = -1 < 0, \quad g_3(p_1) = -1.5 < 0, \quad g_4(p_1) = -1.5 < 0, \quad g_5(p_1) = 0.$$
Active set $I(p_1) = \{1, 5\}$, so set $\mu_2 = \mu_3 = \mu_4 = 0$. Stationarity becomes
$$\begin{aligned} 3 - \mu_1 = 0 &\implies \mu_1 = 3, \\ 3 - \mu_1 = 0 &\text{ (consistent)}, \\ 4 - \mu_1 - \mu_5 = 0 &\implies \mu_5 = 1. \end{aligned}$$
Thus a feasible multiplier vector is $\mu = (3, 0, 0, 0, 1)$, which is nonnegative and satisfies complementary slackness.

We have $\nabla g_1(p_1) = (-1, -1, -1)^\top$, $\nabla g_5(p_1) = (0, 0, -1)^\top$. Since $\nabla g_1(p_1)$ and $\nabla g_5(p_1)$ are linearly independent, p_1 is regular point.

Second-order condition: $\nabla_x^2 \ell(p_1) = \nabla^2 f = 2I_3$. The tangent space to active constraints is
$$T = \{d \in \mathbb{R}^3 : -d_1 - d_2 - d_3 = 0,\ -d_3 = 0\} = \{(t, -t, 0) : t \in \mathbb{R}\}.$$
For $d = (t, -t, 0) \neq 0$,
$$d^\top (2I_3) d = 2(t^2 + t^2) = 4t^2 > 0,$$
so $\nabla_x^2 \ell(p_1)$ is positive definite on T. Thus the second-order sufficient condition holds and p_1 is a strict local minimizer.

Numeric objective: $f(p_1) = 2.25 + 2.25 + 4 = 8.5$.

Check for $p_2 = (4/3, 0.5, 3)$. Compute

$$x_1 + x_2 + x_3 = \tfrac{4}{3} + 0.5 + 3 = \tfrac{29}{6} \approx 4.8333, \quad g_1(p_2) = 5 - \tfrac{29}{6} = \tfrac{1}{6} > 0,$$

and

$$g_2(p_2) = 2 - (0.5 \cdot 3) = 0.5 > 0.$$

Thus p_2 violates g_1 and g_2 and is not feasible. Therefore p_2 cannot satisfy KKT.

Check for $p_3 = (2, 1, 2)$. Evaluate:

$$g_1(p_3) = 0, \quad g_2(p_3) = 0, \quad g_3(p_3) = -2 < 0, \quad g_4(p_3) = -1 < 0, \quad g_5(p_3) = 0.$$

Active set $\mathcal{A} = \{1, 2, 5\}$, hence $\mu_3 = \mu_4 = 0$. Stationarity gives

$$\begin{aligned} 4 - \mu_1 = 0 &\implies \mu_1 = 4, \\ 2 - 4 - 2\mu_2 = 0 &\implies \mu_2 = -1, \\ 4 - 4 - \mu_2 - \mu_5 = 0 &\implies -\mu_2 - \mu_5 = 0. \end{aligned}$$

But $\mu_2 = -1 < 0$ violates dual feasibility. Therefore no nonnegative multipliers exist and p_3 does not satisfy KKT.

Numeric objective: $f(p_3) = 4 + 1 + 4 = 9$.

Final remarks. Among the three candidates, p_1 is the only one that satisfies all KKT conditions and second-order sufficiency; p_2 is infeasible; p_3 is feasible but not a KKT point.

$\Diamond$

Example 5.7.12. A real estate company wants to construct a multistory apartment building on a 500×500 ft lot. The requirements are as follows:

- The total floor space must be $F = 8 \times 10^5$ ft^2.
- The height of each story is 12 ft.
- The maximum building height is restricted to 75 ft.
- The parking area must be at least 10% of the total floor area: $P \geq 0.1F$.

The cost of the building is estimated as

$$C(h, F, P) = 500{,}000h + 2000F + 500P,$$

where

- $h =$ height of the building (in ft),
- $F =$ total floor area (in ft^2),
- $P =$ parking area (in ft^2).

Problem data

Lot area: $500 \times 500 = 250{,}000$ ft^2.
Required total floor area: $F = 8 \times 10^5 = 800{,}000$ ft^2.
Story height $= 12$ ft. Max height $h \leq 75$ ft.
Parking required $P \geq 0.1F$.
Cost: $C(h, F, P) = 500{,}000h + 2000F + 500P$.

Footprint $A = \dfrac{F}{\#\text{stories}} = \dfrac{12F}{h}$.

Optimization variables and constraints

Since F is fixed, minimize (drop constant term $2000F$)

$$\varphi(h, P) = 500{,}000\, h + 500\, P$$

subject to

$$\begin{aligned} g_1(h, P) &:= 0.1F - P \leq 0, \\ g_2(h, P) &:= \frac{12F}{h} + P - A_{\text{lot}} \leq 0, \\ g_3(h) &:= h - 75 \leq 0, \qquad h > 0,\ P \geq 0. \end{aligned}$$

Numerical constants: $F = 800{,}000,\ 0.1F = 80{,}000,\ 12F = 9{,}600{,}000,\ A_{\text{lot}} = 250{,}000.$

Lagrangian and stationarity

Lagrangian with multipliers $\lambda_i \geq 0$ for g_i:

$$\mathcal{L} = 500{,}000h + 500P + \lambda_1(0.1F - P) + \lambda_2\Big(\frac{12F}{h} + P - A_{\text{lot}}\Big) + \lambda_3(h - 75).$$

Stationarity:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial h} &:\ 500{,}000 - \lambda_2 \frac{12F}{h^2} + \lambda_3 = 0, \\ \frac{\partial \mathcal{L}}{\partial P} &:\ 500 - \lambda_1 + \lambda_2 = 0. \end{aligned}$$

Feasibility lower bound for h

To have feasibility with the minimum parking $P = 0.1F$, require

$$\frac{12F}{h} + 0.1F \leq A_{\text{lot}} \quad \Longrightarrow \quad h \geq \frac{12F}{A_{\text{lot}} - 0.1F}.$$

Numerically,

$$h \geq \frac{12 \cdot 800{,}000}{250{,}000 - 80{,}000} = \frac{9{,}600{,}000}{170{,}000} = \frac{960}{17} \approx 56.470588 \text{ ft}.$$

Active set guess and solution

Because the cost increases with h and P, the minimizer will use the smallest feasible values. Thus guess both lower bounds active:

$$P = 0.1F, \qquad \frac{12F}{h} + P = A_{\text{lot}}, \qquad h < 75.$$

Solving gives

$$\boxed{h^* = \frac{12F}{A_{\text{lot}} - 0.1F} = \frac{960}{17}\text{ ft} \approx 56.470588\text{ ft},}$$

$$\boxed{P^* = 0.1F = 80{,}000\text{ ft}^2, \qquad A^* = 170{,}000\text{ ft}^2.}$$

From stationarity with $\lambda_3 = 0$:

$$\lambda_2 = 500{,}000\frac{(h^*)^2}{12F}, \qquad \lambda_1 = 500 + \lambda_2.$$

Numerically $\lambda_2 \approx 166.43$, $\lambda_1 \approx 666.43$, both nonnegative, consistent with the active set.

Second-order condition

The Lagrangian Hessian (in (h, P)) has

$$\frac{\partial^2 \mathcal{L}}{\partial h^2} = 24\lambda_2\frac{F}{h^3}, \quad \frac{\partial^2 \mathcal{L}}{\partial P^2} = 0, \quad \frac{\partial^2 \mathcal{L}}{\partial h \partial P} = 0.$$

The tangent space for the active equalities (linearized) is trivial (only zero vector), hence the SOSC holds and the point is a strict local minimizer. In this setting this is the global minimizer for the continuous model.

Full cost at optimum

Adding back the constant $2000F$,

$$\begin{aligned} C^* &= 500{,}000h^* + 2000F + 500P^* \\ &= 500{,}000 \cdot \frac{960}{17} + 2000 \cdot 800{,}000 + 500 \cdot 80{,}000 \\ &= \frac{480{,}000{,}000}{17} + 1{,}600{,}000{,}000 + 40{,}000{,}000 \\ &\approx 1{,}668{,}235{,}294.12 \text{ (dollars).} \end{aligned}$$

Conclusion

The continuous-model optimal design is:

$$\boxed{h^* = \frac{960}{17}\text{ ft} \approx 56.4706\text{ ft}, \quad P^* = 80{,}000\text{ ft}^2, \quad A^* = 170{,}000\text{ ft}^2,}$$

with corresponding cost $\approx \$1.668 \times 10^9$. All KKT conditions are satisfied and the second-order sufficient condition holds.

$\diamondsuit$

5.8 Lagrange Duality

Given a non linear constrained optimization problem, known as primal, there exists another non linear optimization problem closely related to it which is called the Lagrangian dual of the primal. Under certain assumptions, the primal and its dual have same optimal objective function values.

Consider the problem

$$\boxed{\begin{array}{r} \text{minimize } f(x) \\ \text{subjected to} \\ h_1(x) = 0, \ldots, h_m(x) = 0 \\ g_1(x) \leq 0, \ldots, g_p(x) \leq 0 \end{array}} \tag{5.41}$$

Let $\Omega = \{x \in \mathbb{R}^n : h(x) = 0, g_j(x) \leq 0, 1 \leq i \leq m, 1 \leq j \leq p\}$.

As before, we can write the above problem as

$$\boxed{\min f(x) \text{ subjected to } h(x) = 0 \text{ and } g(x) \leq 0} \tag{5.42}$$

where $f\colon \mathbb{R}^n \to \mathbb{R}, g\colon \mathbb{R}^n \to \mathbb{R}^p, h\colon \mathbb{R}^n \to \mathbb{R}^m$. The problem (5.42) is called the *primal problem.*

The Lagrangian function associated with the problem (5.42) is a function $\ell\colon \Omega \times \mathbb{R}^m \times \mathbb{R}^p \to \mathbb{R}$ given by

$$\ell(x, \lambda, \mu) = f(x) + \sum_{i=1}^{m} \lambda_i h_i(x) + \sum_{j=1}^{p} \mu_j g_j(x). \tag{5.43}$$

The *Lagrange relaxation function* of the primal (5.42) is given by

$$\tilde{\ell}(\lambda, \mu) = \inf_{x \in \Omega} \ell(x, \lambda, \mu). \tag{5.44}$$

Lemma 5.8.1. *If p^* is an optimal value of the problem 5.42, then we have*

$$\tilde{\ell}(\lambda, \mu) \leq p^* \textit{ for all } \lambda \in \mathbb{R}^m \textit{ and } 0 \leq \mu \in \mathbb{R}^p.$$

Proof. Since $\mu \geq 0$ and $g \leq 0$, we have

$$\ell(x, \lambda, \mu) = f(x) + \sum_{i=1}^{m} \lambda_i h_i(x) + \sum_{j=1}^{p} \mu_j g_j(x) \leq f(x) \text{ for all } x \in \mathbb{R}^n.$$

Thus

$$\tilde{\ell}(\lambda, \mu) = \inf_{x \in \Omega} \tilde{\ell}(\lambda, \mu) \leq f(x) \text{ for all } x \in \mathbb{R}^n.$$

Hence

$$\tilde{\ell}(\lambda, \mu) \leq \inf_{x} f(x) = f(x^*).$$

□

Lemma 5.8.2. *The Lagrange relaxation function $\tilde{\ell}(\lambda, \mu)$ is a concave function.*

Proof. Letf as an exercise. □

The corresponding **dual problem** is given by

$$\boxed{\max_{\lambda \in \mathbb{R}^m, \mu \in \mathbb{R}^p} \tilde{\ell}(\lambda, \mu) \text{ subjected to } \mu \geq 0.} \tag{5.45}$$

The optimal value d^* is

$$d^* = \max_{\lambda \in \mathbb{R}^m, 0 \leq \mu \in \mathbb{R}^p} \tilde{\ell}(\lambda, \mu)$$

The difference $p^* - d^*$ is called the **duality gap.**

Example 5.8.3. Consider a minimization problem

$$\text{minimize } f(x) \quad \text{subjected to } g(x) \leq 0$$

where $f, g\colon \mathbb{R}^n \to \mathbb{R}$. Let the feasible set for this problem be S. Define the set

$$G = \{(y, z) : y = g(x), z = f(x) \text{ for some } x \in S\}.$$

Note that G is the image of S under the map (g, f). See Fig. 5.8.

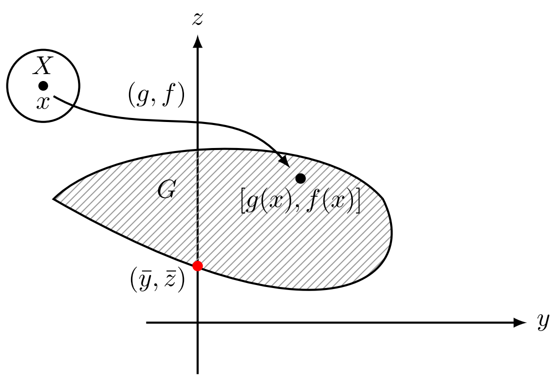

Figure 5.8: Figure 1 for Example 5.8.3

Then the primal is to find the point (y, z) in G with $y \leq 0$ and that has minimum ordinate z. Let this point be denoted by $(\overline{y}, \overline{z})$. See Fig. 5.8.

The Lagrangian dual problem is

$$\text{minimize } \theta(\mu) \quad \text{subjected to } \mu \geq 0$$

where

$$\theta(\mu) = \inf_{x \in S} \{f(x) + \mu g(x)\}.$$

Given $\mu \geq 0$, the given the primal, its Lagrangian relaxation function is equivalent to minimizing $z + \mu y$ over the set of points $(y, z) \in G$. Notice that for $\theta(\mu) = \alpha$, $z + \mu y = \alpha$ is the line with slope $-\mu$ and z-intercept α.

In order to minimize $z + \mu y$, over G, we need to translate the $z + \mu y = \alpha$ parallel to itself as far as possible keeping it in contact with G. The last intercept on the z-axis is the value of $\theta(\mu)$ corresponding to the given $\mu \geq 0$. See Fig. 5.9.

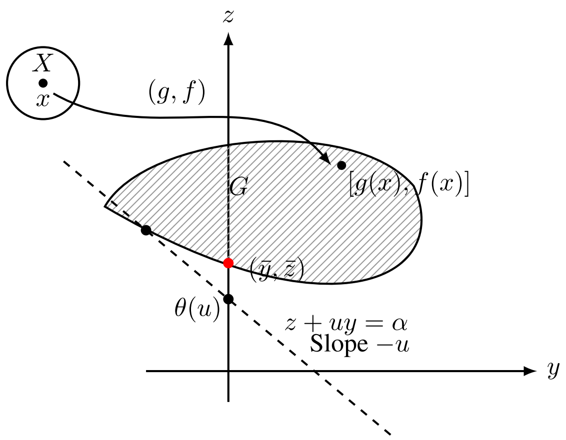

Figure 5.9: Figure 2 for Example 5.8.3

Finally to solve the dual problem, we need to find the line with slope $-\mu$ ($\mu \geq 0$) such that the last intercept on z-axis, $\theta(\mu)$ is maximum. Such a line has slope $-\mu^*$ and supports the set G at the point $(\overline{z}, \overline{y})$. Hence the solution of the dual problem is $\overline{\mu}$ and the optimal value is $\overline{y}$. See Fig. 5.10.

Note that the solution of the primal and dual both are $\overline{z}$.

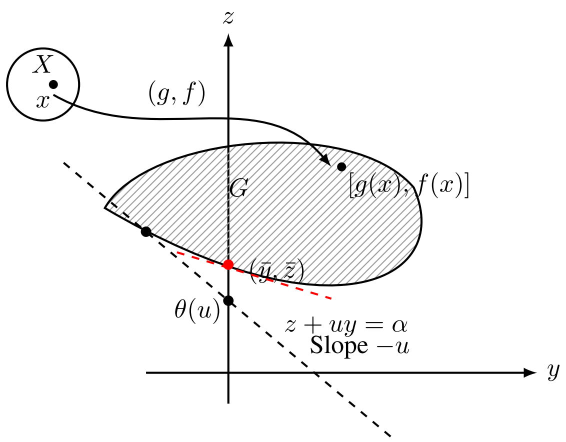

Figure 5.10: Figure 3 for Example 5.8.3

$\Diamond$

Example 5.8.4. Consider a linear programming problem

$$\left.\begin{array}{c}\text{minimize } c^T x \\ \text{subjected to } Ax \geq b\end{array}\right\} \tag{5.46}$$

The Lagrangian function of (5.46) is given by

$$\ell(x, \mu) = c^T x + \mu^T (b - Ax) = \mu^T b + (c^T - \mu^T A)x.$$

The dual function is given by

$$\varphi(\mu) = \inf_{x \in \mathbb{R}^n} \ell(x, \mu) = \begin{cases} -\infty & \text{if } c^T - \mu^T A \neq 0 \\ \mu^T b & \text{if } c^T - \mu^T A = 0. \end{cases}$$

Hence the dual programme of the problem (5.46) is given by

$$\sup_{\mu \geq 0} \varphi(\mu) \to \begin{cases} \max \mu^T b \\ \mu^T A = c^T \\ \mu \geq 0 \end{cases}$$

◊

Example 5.8.5. Consider a linear programming problem

$$\left.\begin{aligned} \text{minimize } c^T x \\ \text{subjected to } Ax \geq b \\ x \geq 0. \end{aligned}\right\} \tag{5.47}$$

Note that the two constraints can be combined to one as

$$\begin{bmatrix} A \\ -- \\ I_{n\times n} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \geq \begin{bmatrix} b \\ -- \\ 0_{n\times 1} \end{bmatrix}.$$

Let $\tilde{A} = \begin{bmatrix} A \\ -- \\ I_{n\times n} \end{bmatrix}$ and $\tilde{b} = \begin{bmatrix} b \\ -- \\ 0_{n\times 1} \end{bmatrix}$.

The Lagrangian function of (5.47) is given by

$$\ell(x, \tilde{\mu}) = c^T x + \tilde{\mu}^T(\tilde{b} - \tilde{A}x) = \tilde{\mu}^T \tilde{b} + (c^T - \tilde{\mu}^T \tilde{A})x,$$

where $\tilde{\mu} = (\mu_1, \cdots, \mu_m, \mu_{m+1}, \cdots, \mu_{m+n}) \in \mathbb{R}^{m+n}$.

The dual programme of the problem (5.46) is given by

$$\text{maximize } \tilde{\mu}^T \tilde{b}, \qquad \text{subjected to } \tilde{\mu}^T \tilde{A} = c^T, \text{ and } \tilde{\mu} \geq 0.$$

Note that

$$\tilde{\mu}^T A = c^T \implies \begin{bmatrix} \mu_1 & \cdots & \mu_m \end{bmatrix} A + \begin{bmatrix} \mu_{m+1} & \cdots & \mu_{m+n} \end{bmatrix} I_{n\times n} = c^T.$$

This implies $\begin{bmatrix} \mu_1 & \cdots & \mu_m \end{bmatrix} A \leq c^T$, since $\mu_{m+1}, \cdots, \mu_{m+n} \geq 0$.

Thus the dual of the primal is given by

$$\left.\begin{aligned} \max \mu^T b \\ \text{subjected to } A^T \mu \leq c \\ \mu \geq 0. \end{aligned}\right\}$$

◊

Example 5.8.6. Consider the problem

$$\left.\begin{aligned} \text{minimize } \ ||x||^2 \\ \text{subject to } Ax = b \end{aligned}\right\} \tag{5.48}$$

where $A \in \mathbb{R}^{m\times n}$ a full rank matrix. The Lagrangian is given by

$$\ell(x, \lambda) = x^T x + \lambda^T (Ax - b)$$

In order to find the Lagrange relaxation function, we find the gradient of the Lagrangian and equate to zero. Thus we have

$$\lambda \ell(x, \lambda) = 2x + A^T \lambda = 0, \implies x = -\frac{1}{2} A^T \lambda.$$

Hence the Lagrange relaxation function is given by

$$\varphi(\lambda) = \inf \ell(x, \lambda) = -\frac{1}{4}\lambda^T(AA^T)\lambda - \lambda^T b.$$

Hence the Lagrange dual is given by

$$\max_{\lambda} -\frac{1}{4}\lambda^T(AA^T)\lambda - \lambda^T b$$

which is an unconstrained concave quadratic maximization problem.

This can be solve analytically and easy to see that the the solution occurs at

$$\lambda^* = -2(AA^T)^{-1}b$$

with dual optimal value

$$d^* = b^T(AA^T)^{-1}b$$

◊

Weak Duality Theorem

Let the optimal value of the primal (5.42) is $\nu(P)$. Then we have the following result which is know as *weak duality.*

Theorem 5.8.1. *Consider the primal* (5.42) *and its dual* (5.45)*. Let x be a feasible solution to the primal* (5.42) *(i.e. $h(x) = 0, g(x) \leq 0$) and (λ, μ) a feasible solution of the dual* (5.45) *(i.e, $\mu \geq 0$). Then*

$$f(x) \geq \tilde{\ell}(\lambda, \mu).$$

Proof. Since x is a feasible solution and $\mu \geq 0$, we have $h(x) = 0$ and $\mu_i g_i(x) \leq 0$. Hence

$$\tilde{\ell}(\lambda, \mu) = \inf \left\{ f(x) + \sum_{i=1}^{m} \lambda_i h_i(x) + \sum_{j=1}^{p} \mu_j g_j(x) \right\} \leq f(x).$$

□

Corollary 5.8.7.

$$\nu(P) = \inf_{\{x\in\Omega : g(x)\leq 0, h(x)=0\}} \{f(x)\} \geq \sup_{(\lambda,\mu)} \{\tilde{\ell}(\lambda, \mu) : \mu \geq 0\}.$$

The Corollary says that the optimal objective function value of the primal is greater than or equal to the optimal objective function value of the corresponding dual. If there is a strict inequality in the Cor. 5.8.7, then we say that there exists *duality gap.*

Strong Duality

We show that under certain conditions, there is no duality gap. This is what is called strong duality theorem.

Theorem 5.8.2 (Strong Duality Theorem)**.** *Let $X \subset \mathbb{R}^n$ be a convex set. Let $f\colon \mathbb{R}^n \to \mathbb{R}$, $g\colon \mathbb{R}^n \to \mathbb{R}^p$ be convex functions and $h\colon \mathbb{R}^n \to \mathbb{R}^m$ be affine map.. Suppose the following constrained qualification is satisfied. There exists $x^* \in X$, such that $g(x^*) < 0$, $h(x^*) = 0$ and $0 \in \operatorname{int} h(X)$, then*

$$\inf\{f(x) : x \in X, g(x) \leq 0, h(x) = 0\} = \sup\{\tilde{\ell}(\lambda, \mu) : \mu \geq 0\}$$

where $\tilde{\ell}(\lambda, \mu) = \inf\{f(x) + \lambda^T h(x) + \mu^T g(x) : x \in X\}$. Furthermore, if the $\inf$ *is finite, then* $\sup\{\tilde{\ell}(\lambda, \mu) : \mu \geq 0\}$ *is achieved at a point (λ^*, μ^*) with $\mu^* \geq 0$. If the* $\inf$ *is achieved at x^*, then $\mu^T g(x^*) = 0$.*

5.9 Exercises on KKT

1. Consider a matrix $L = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$. Check if L is negative definite on the space $W = \{(x_1, x_2, x_3) \in \mathbb{R}^3 : x_1 + x_2 + x_3 = 0\}$.

2. Consider a matrix $L = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 3 \end{pmatrix}$. Check if L is negative definite on the space $W = \{(x_1, x_2, x_3) \in \mathbb{R}^3 : x_1 = 0\}$.

3. Minimize $f(x_1, x_2) = 2x_1^2 + 2x_1x_2 + x_2^2 - 10x_1 - 10x_2$ subjected to $x_1^2 + x_2^2 \leq 5$ and $3x_1 + x_2 \leq 6$. Use the second order sufficient conditions to justify your answer.

4. Minimize $f(x, y) = 2x^2 + 3y^2$ subjected to $3 - x - y^2/4 >= 0, x - y >= 1, x + y >= 1$.

5. Minimize $x_1^2 + x_2^2$ Subjected to $4 - x_1 - x_2^2 \leq 0$, $3x_2 - x_1 \leq 0$ and $-3x_2 - x_1 \leq 0$. Use the second order sufficient conditions to justify your answer.

6. Let A be $m \times n$ full rank real matrix with, $m < n$. Consider the minimization problem

$$\text{minimize } c^T x$$
$$\text{Subjected to } Ax \leq 0$$

Using KKT method show that if there exists a solution then the value of objective function must be 0.

7. Let P be an $n\times$ positive definite real symmetric matrix and A, $m \times n$ matrix. Let $b \in \mathbb{R}^m$ with $b \geq 0$. Find all the KKT points of the problem

$$\text{minimize } x^T P x$$
$$\text{Subjected to } Ax \leq b$$

8. Find a minimum of $f(x_1, x_2) = -x_1 - 3x_2$ subjected to
$x_1 + x_2 = 6 \;\; \& \; -x_1 + x_2 \leq 4$ using the KKT conditions.

$x = 1, y = 5$ and $f = -16$

9. Find a maximum of $f(x_1, x_2) = x_1 + x_2$ subjected to
$x_1 \leq 2 \;\; \& \; x_1 + x_2^2 \leq 5$ using the KKT conditions.

Sol: $x_1 = 2.0, x_2 = -1.73205$ and $f = -3.73205$

10. Find a minimum of $f(x_1, x_2) = 6x_1 + x_2$ subjected to
$2x_1 + 7x_2 \geq 3 \;\; \& \; 2x_1 - x_2 \geq 2$ using the KKT conditions.

Sol: $x = 1.0625, y = 0.125$ and $f = 6.5$.

11. Find a minimum of $f(x_1, x_2) = -6x_1 + 9x_2$ subjected to
$x_1 - x_2 \geq 2,\;\; 3x_1 + x_2 \geq 1,\;\; \& \; 2x_1 - 3x_2 \geq 3$ using the KKT conditions.

No valid solution

12. Find a minimum of $f(x_1, x_2) = \dfrac{1}{x_1^2 + x_2^2}$ subjected to
$x_1^2 + x_2^2 \leq 5,\; x_1 \leq 10,\; x_2 \leq 4 \;\; \& \; x_1 + 2x_2 = 4$ using the KKT conditions.

Ans: There are three valid KKT points:
(i) $x = -0.4, y = 2.2$, (ii) $x = 0.8, y = 1.6$ and (iii) $x = 2, y = 1$